Braille → print, two ways

With and without braille-native tokens: a 135M-parameter evaluation on 120 held-out sentences, plus a 120-row kit for comparing another LLM.

Same base weights, same training sentences, same optimizer and seed. The only intended difference is how braille is presented to the model.

MODEL	CORPUS	HARDWARE	EXPERIMENT
SmolLM2-135M	600 sentences · 138 words	Snapdragon X1 · 16 GB · no CUDA	2 arms · 4 runs

GO
SCOPED FOLLOW-UP

VERDICT

The braille-native representation works: **120 / 120 exact translations** against **0 / 120** for the matched baseline. A 256-token control, run after the main comparison, also scores 0 / 120 — so the baseline's collapse is not a truncation artefact. This justifies a larger study, starting with a lexically independent test set. The result is evidence about representation, not yet about open-domain braille translation.

EXACT MATCH	CHARACTER ERROR RATE	SACREBLEU
120 / 120	**0.0%**	**100.0**
baseline **0 / 120** · +100 pts	baseline **63.8%** · mean per sentence	baseline **4.0** · corpus, n=120

INPUT TOKENS PER CELL	DISTINCT OUTPUTS	BATCHED LATENCY
0.535	**120**	**0.23 s**
baseline **3.00** · **5.6×** fewer	baseline **9** · for 120 inputs	baseline **0.49 s** · per sentence, batch 8

ABSTRACT

A braille→print proof of concept was trained twice on the same 480 sentences. The **baseline** read raw Unicode braille through the stock byte-level tokenizer. The **native** model added 177 braille-cell tokens, initialised their embeddings from the print strings they represent, and fine-tuned LoRA adapters. On the 120 held-out sentences, native translated every sentence exactly. Baseline translated none: it emitted nine distinct sentences, used 22 of the 138 corpus words, and matched the actor noun in 2.7% of cases. The comparison isolates tokenisation as the decisive variable at this scale; the corpus design limits what can be claimed beyond it. Sections 12–21 add the model and data cards, every test item, the comparison kit, and Appendix C.

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01 Executive summary

The project asked a narrow question: at fixed model size and data, does a braille-native tokenizer make braille→print translation work where the stock tokenizer does not? The answer on this corpus is unambiguous: yes, and the gap is qualitative, not incremental.

Two arms were trained from `HuggingFaceTB/SmolLM2-135M` on the same 480 UEB Grade 1 sentences: identical seed, one epoch, AdamW at 1e-3, batch 4, fp32 on CPU. The baseline fed raw Unicode braille to the stock tokenizer. The native arm added **177 tokens** (27 corpus cells + 150 BPE merges), set each new embedding to the mean of the print-string embeddings it expands to, and trained LoRA adapters plus the resized embedding matrix and LM head. Evaluation used 120 held-out sentences, greedy decoding, and exact match as the primary metric.

Native produced **120 distinct, exact sentences** (CER 0.0, BLEU 100.0). Baseline produced **nine distinct sentences for 120 inputs** (CER 63.8%, BLEU 4.0). The baseline did not fail by producing noisy translations; it failed by not conditioning on the input at all. All 120 outputs begin with “The” although 30 references begin with “A” and 3 with “My”. Its output vocabulary is 22 words, and 116 of the 138 reference words are never emitted. On the nine-word template it matches the final article 100% of the time and the actor noun 2.7% of the time. That is template completion, not decoding.

The mechanism is plausible and consistent with the measurements. Raw braille costs **3.00 tokens per cell** (165 tokens per sentence), so the model must learn a mapping through a fragmented byte-level representation. Native costs **0.535 tokens per cell** (29 tokens per sentence), with embeddings that already point at the right print fragments. Training loss separates the arms early: native falls below 0.3 by step 30 and reaches 0.002; baseline is still at 1.32 at step 120 and never converges.

KEY FINDINGS

01 **Representation was the bottleneck.** Same weights, data, optimizer, and seed. The only intended change was tokenizer plus embedding initialisation, and exact match moved from 0% to 100%.

02 **The baseline’s failure is structural.** Nine outputs, 2.62 bits of output entropy (maximum 6.91), 22-word vocabulary, and near-zero accuracy on every content slot. It learned the corpus prior, not the dots.

03 **Efficiency is a second, independent gain.** 5.6× fewer input tokens, 2.2× faster batched inference, and a prompt that fits the 128-token training budget that the baseline overruns on every sentence.

04 **The corpus is the ceiling on the claim.** 138 words, three sentence patterns, and every test word seen in training. Native’s 120/120 demonstrates learnability on this distribution; it does not demonstrate generalisation.

05 **The truncation confound was tested and dismissed.** The baseline lost an average of 8.5 tokens from 61% of its training prompts at 192 tokens. A control at 256 tokens — where no row is truncated — still scores **0 / 120** exact, CER 60.2%, BLEU 4.2, with 15 distinct outputs. More context slightly diversified the prior; it did not create input dependence.

> WHAT WOULD OVERTURN THE VERDICT
>
> A word- or template-disjoint test set on which native drops sharply; or a probe showing that native copies template endings rather than reading cells. Neither has been run. Both are P0 in §10. The 256-token control has now run and did not overturn the result. Until the two remaining tests pass, the defensible claim is: **the representation works on this corpus**.

> READING THIS REPORT
>
> §2–§5 define the experiment and its limits. §6–§8 present outcomes and diagnose the baseline. §9 lists every threat we know of, with direction and mitigation. §10 is the decision and the next plan. §11 reproduces the run. §12–§13 are model and data cards, §14–§17 print the full held-out set, and §18–§19 are the comparison kit: exact prompt, conditions, scorer, and score sheet.

02 Question and design

A MATCHED TWO-ARM COMPARISON

Hypothesis: at fixed model capacity and training data, braille→print quality is limited primarily by how braille is represented to the model, not by the model's size.

If true, a small base model should learn the mapping when each braille cell costs about one token and its embedding starts near the print fragment it encodes; the same model should fail when the same dots are spread across three byte-level tokens with no print-aligned initialisation. The experiment changes only that representation.

TABLE 1 — THE TWO ARMS

PROPERTY	BASELINE	NATIVE
Base model	`SmolLM2-135M`	`SmolLM2-135M`
Braille tokenizer	Stock byte-level BPE (49,152)	+177 braille tokens (49,329)
New-token init	—	BKFT: mean of print-string embeddings
Trainable parameters	460,800 (0.34%)	57.3M reported (29.85%, incl. saved embeddings)
Max sequence length	192 tokens	128 tokens
Training data	480 sentences	480 sentences
Optimizer / schedule	AdamW · lr 1e-3 · 1 epoch · 120 steps	AdamW · lr 1e-3 · 1 epoch · 120 steps
Batch · precision · seed	4 · fp32 CPU · 0	4 · fp32 CPU · 0

The intended variables are the tokenizer, the embedding initialisation, and the parameters that consequently become trainable. The `max_len` difference is a consequence of the representation, not a tuning choice: native fits comfortably in 128 tokens, while baseline needs 165 tokens for braille alone. At 192, every answer survived intact, but 61% of training rows still lost prompt tokens from the left (§9).

EVALUATION PROTOCOL

- **Exact match (primary).** The whole output must equal the reference after whitespace trimming. It is insensitive to BLEU's behaviour on short sentences and is the metric a user experiences.
- **Character error rate.** Mean Levenshtein distance divided by reference length, per sentence. It measures severity when outputs are wrong.
- **sacreBLEU.** Corpus score with the default tokenizer. Reported alongside its four n-gram precisions and brevity penalty so that template matching is visible.
- **Tokens per cell.** Braille-side token count under each model's own tokenizer, divided by the number of Unicode braille cells. This is the efficiency claim.
- **Output diversity.** Distinct outputs, Shannon entropy, largest cluster, and output-vocabulary coverage. These expose input bypass, which CER and BLEU alone can hide.
- **Position accuracy.** Per-word-slot match on the 112 nine-word test sentences. It localises where a model succeeds and fails inside the template.
- **Latency.** Greedy decoding on eight test sentences after one warm-up, measured at batch 1 and batch 8 on the same machine.

> OUT OF SCOPE
> Grade 2 contractions; digits and punctuation beyond the period; line breaks and multi-sentence documents; novel vocabulary; other model sizes; GPU or bfloat16; sampling; human evaluation; production throughput. Any of these could change the measured gap.

03 Corpus

600 TEMPLATED SENTENCES, CLOSED VOCABULARY

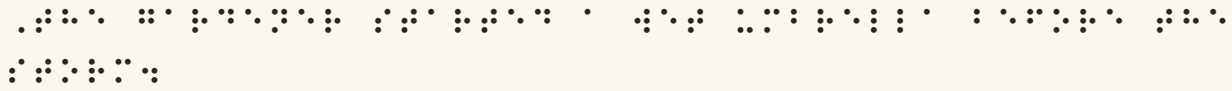
The corpus is a controlled probe, not a sample of braille reading. It was built from `sentences.txt`, converted to UEB Grade 1 with the project's built-in engine (liblouis was not installed), filtered by round-trip equality, and split 80/20 with seed 0.

TABLE 2 — CORPUS PROPERTIES

PROPERTY	VALUE	NOTE
Unique sentences	600	generated from a fixed bank of actors, verbs, objects, and places
Train / test	480 / 120	random split; no braille or print string shared
Vocabulary	138 words	closed; all 138 test words occur in training
Length	9.13 words	525 × 9, 72 × 10, 3 × 11; mean 54.3 characters
Sentence patterns	3	article–actor–verb–object–place, with local variants
Character set	A–Z, space, “.”	capital letters only at sentence start
Braille length	55.3 cells	UEB Grade 1, uncontracted
Round-trip filter	600 / 600	braille → print reproduces the source exactly

Test vocabulary coverage is 138 / 138. There are no out-of-vocabulary words in the evaluation.

SPECIMEN · TEST ROW 01



The gardener started a wet umbrella before the storm.

54 CELLS · 162 BASELINE TOKENS · 24 NATIVE TOKENS

The round-trip filter guarantees clean labels: every kept pair is one the engine can invert, so accuracy differences cannot come from annotation noise. The price is selection bias toward strings the engine handles perfectly — which, for this corpus, is all of them. The shortcut is elsewhere: a random split shares words and sentence patterns between train and test. A model can score well by learning which word belongs in each slot and copying the rest.

That is not a hypothetical. It is precisely the baseline's behaviour (§8): it fills the template with high-frequency fragments and ignores the input. Native's 120/120 therefore establishes that the representation is sufficient to learn this distribution. It does not establish that the system reads arbitrary braille, and it cannot, because the test set contains no arbitrary braille.

> WHY “BIGGER” IS NOT THE NEXT STEP
>
> Expanding the same generator to 1,500–2,000 sentences would add more combinations of the same 138 words and three patterns. It would lower training loss and raise confidence on this distribution, but it would not test generalisation. The next corpus needs new words, new syntax, and a split that holds out words or patterns — otherwise more data measures memorisation more precisely.

4.1 TWO TOKENISATIONS OF THE SAME DOTS

The baseline sends the braille string as Unicode text. SmolLM2's byte-level BPE has no braille-aware units, so a cell shatters into roughly three tokens: **3.00 tokens per cell**, 165.3 tokens per sentence. The native arm starts from the 27 distinct cells the corpus uses (letters, space, capital sign, period) and learns 150 BPE merges over training braille, never merging across spaces. The result is 177 added tokens, a vocabulary of 49,329, and **0.535 tokens per cell** — 1.87 cells per token, 29.4 tokens per sentence.

TABLE 3 — INPUT COST

	BASELINE	NATIVE
Vocabulary	49,152	49,329
Tokens / cell	3.000	0.535
Tokens / sentence	165.3	29.4
Cells / token	0.33	1.87
Test prompts over 128 tokens	120 / 120	0 / 120

TABLE 4 — LEARNED MERGES

TOKEN	PRINT	COUNT
⠓⠑	he	1,159
⠮⠓⠑	the	1,046
⠑⠗	er	523
⠑⠙	ed	443
⠔⠊	in	387

Merges are orthographic fragments, not whole words. Counts are training-corpus occurrences from `braille_tokens.json`.

4.2 BKFT INITIALISATION

Each added token starts at the mean of the base model's embeddings for the print text it expands to. Capital, number, and space indicators expand to nothing, so they do not pollute the average; when the LM head is untied, the same rows are mirrored there. For a token mapping to `the`, the new row is exactly the centroid of the embeddings of the pieces that spell it. The check in the project's wiring test confirms this is numerically exact.

Why it matters: a random new row must be moved into place by gradient descent from 480 examples. A print-aligned row already sits in the region the model uses to represent that text. The experiment does not separate the contribution of the extra tokens from that of the initialisation — they are delivered as one design — but it does show the combined design is sufficient.

4.3 LORA AND TRAINING TARGET

SETTING	VALUE
LoRA rank · alpha · dropout	`8 · 16 · 0.05`
Target modules	`q_proj, v_proj`
Modules saved (native only)	`embed_tokens, lm_head`
Loss mask	answer tokens only; prompt positions are −100
Prompt	`Translate the following braille to English text. / Braille: {brl} / English:`

Small models tie input and output embeddings. Resizing the vocabulary therefore resizes both, but LoRA alone would leave the new rows frozen; `modules_to_save` makes them trainable. This is why PEFT reports 57.3M trainable parameters (29.85%) for native — most of that is saved embedding copies, not adapter weights. The baseline has 460,800 trainable parameters (0.34%). Examples longer than `max_len` are left-trimmed so the answer is never cut; the consequence for the baseline is quantified in §9.1.

05 Training and compute

ONE EPOCH ON A SNAPDRAGON X1

Both delivered models trained for a single epoch — 120 optimizer steps at batch 4 — on CPU in float32. A three-epoch native plan was stopped after epoch 1: loss was already 0.002 and evaluation exact, so two more passes would have bought nothing.

TABLE 5 — ENVIRONMENT

CPU	Snapdragon X X126100, 8 cores @ 2.96 GHz
Memory	16 GB
OS / Python	Win 11 26300 · 3.12.10 (ARM64)
PyTorch	2.15.0+cpu
Transformers / PEFT	5.22.1 / 0.21.2
CUDA	none

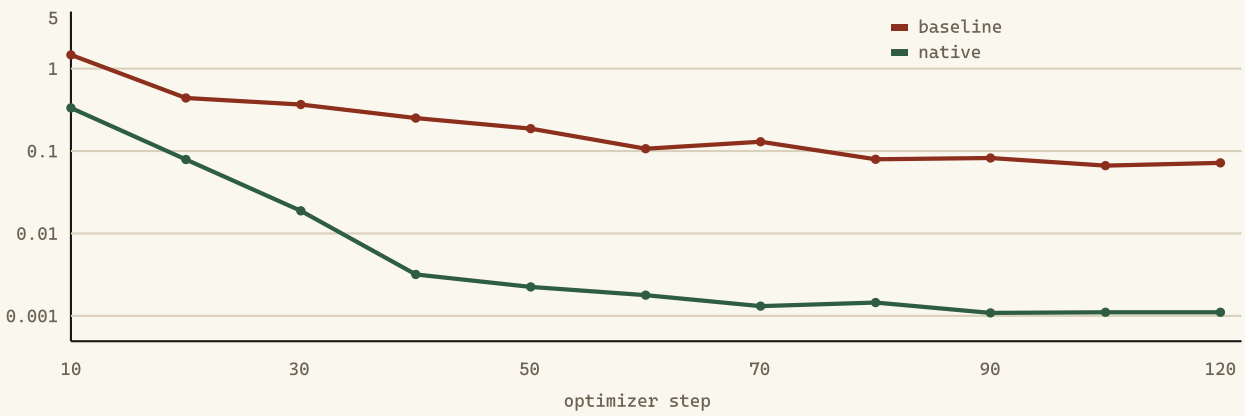
TABLE 6 — MEASURED THROUGHPUT

WORKLOAD	TIME
1536^3 fp32 matmul, 1 thread	45.6 GFLOPS
1536^3 fp32 matmul, 8 threads	223.3 GFLOPS
Native train step (batch 4, 128 tok)	1.94 s
Baseline train step (batch 4, 192 tok)	4.16 s
Native / baseline epoch	≈ 4 / 8 min

Step times measured in isolation with 8 threads; earlier runs shared the machine and were slower. Latency in §6 uses the delivered checkpoints.

5.1 LOSS CURVES

FIG. 1 — TRAINING LOSS, LOG SCALE



Native converges early; baseline plateaus. Native passes below 0.3 at step 30 and reaches 0.002. Baseline descends from 4.2 to ≈1.5 and oscillates, ending at 1.321 (epoch mean 2.421). A model that had learned the mapping would not sit at 1.3 after fitting its own 480 examples.

5.2 EXPERIMENT LOG

RUN	CONFIGURATION	OUTCOME	DECISION
Baseline, first attempt	lr 1e-4 · 1 epoch	One sentence for every input	Discarded: too few effective steps for 460k trainable parameters
Baseline, delivered	lr 1e-3 · max_len 192	Loss 2.421 mean / 1.321 final · 9 outputs	Kept as the control
Native, delivered	lr 1e-3 · max_len 128	Loss 0.501 mean / 0.002 final · 120 exact	Kept as the treatment
Baseline, 256-token control	lr 1e-3 · max_len 256	Loss 2.317 mean · 15 outputs · 0 exact	Truncation confound dismissed

06 Results

120 HELD-OUT SENTENCES, GREEDY

TABLE 7 — HEADLINE METRICS

METRIC	BASELINE	NATIVE	READING
Exact match	0 / 120	120 / 120	every output correct
sacreBLEU	4.0	100.0	corpus score
CER, mean	63.8%	0.0%	median 64.0%, worst 82.2%
Distinct outputs	9	120	of 120 inputs
Output vocabulary	22 / 138	138 / 138	reference words used
Tokens per cell	3.00	0.535	5.6× fewer
Braille tokens / sentence	165.3	29.4	same 55-cell input
Training loss, epoch mean	2.421	0.501	final step 1.32 vs 0.002
Latency, batch 1	0.953 s	0.686 s	per sentence
Latency, batch 8	0.494 s	0.228 s	2.2× faster

The baseline column is the delivered 192-token run. A 256-token control, reported in §9.1, reaches 0 / 120 exact and CER 60.2%.

6.1 WHAT BLEU 4.0 MEANS

BLEU COMPONENT	BASELINE	NATIVE
1-gram precision	33.83	100.0
2-gram precision	3.15	100.0
3-gram precision	2.08	100.0
4-gram precision	1.19	100.0
Brevity penalty	0.993	1.000
Corpus BLEU	4.0	100.0

The baseline's BLEU is not a near-zero accident of length: its brevity penalty is 0.993, so it produces sentences of the right size. The score comes almost entirely from unigrams — chiefly "the" — and disappears by the bigram: 33.8% → 3.15% → 1.19%. Exact match is the cleaner headline; BLEU's component profile shows *why* the baseline looks superficially plausible.

Native is 100 on every component because every sentence is copied exactly. That ceiling is expected and not, by itself, informative about generalisation.

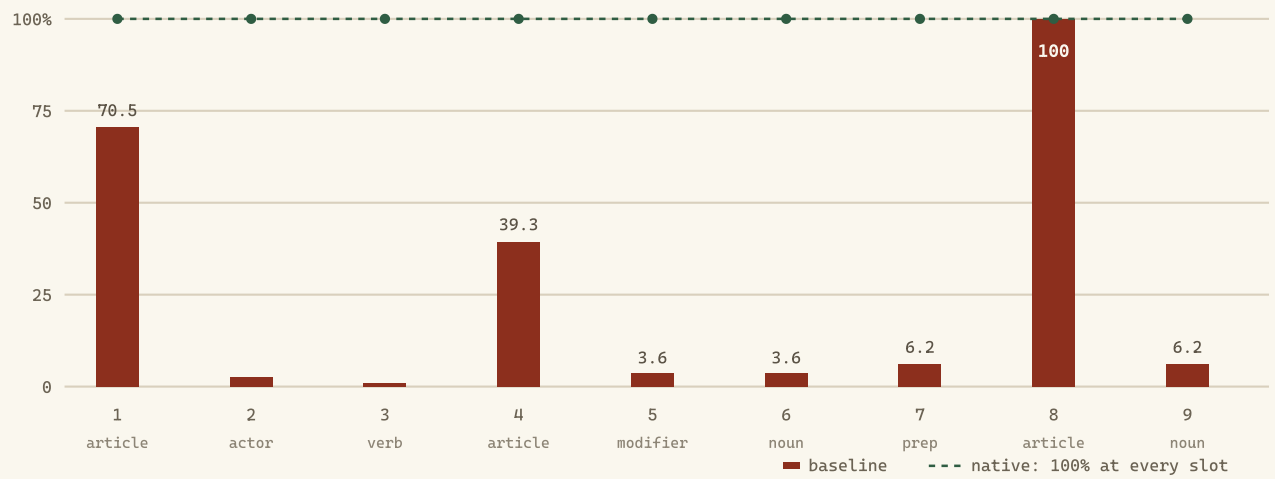
6.2 OUTPUT DIVERSITY

STATISTIC	BASELINE	NATIVE	INTERPRETATION
Distinct outputs	9 (7.5%)	120 (100%)	input-to-output information
Output entropy	2.62 bits	6.91 bits	maximum 6.91 for 120 items
Largest cluster	36 (30%)	1	one sentence covers a third of inputs
Words emitted	22	138	of 138 in the references
Reference words never emitted	116	0	coverage failure, not paraphrase

Nine outputs can carry at most 3.17 bits about the input; the observed distribution carries 2.62. In practice the input determines almost nothing. Native sits at the maximum because every output is different and correct.

> FINDINGS THAT SURVIVE THE METRICS
>
> The baseline's problem is not "low accuracy". It is that the model learned `P(output)` and not `P(output | braille)`. Nine sentences are a prior, not a degraded translation. That distinction is why the native comparison is stronger than a BLEU delta.

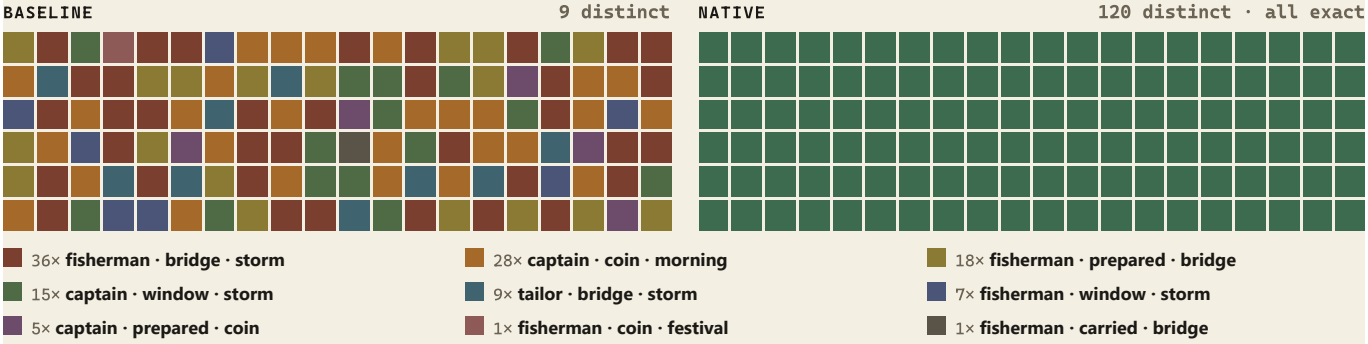
FIG. 2 — WORD ACCURACY BY POSITION · NINE-WORD TEST SENTENCES, N = 112



The shape of a template, not a translation. The baseline matches the final article 100% of the time, the initial article 70.5%, and the second article 39.3% — but the actor noun 2.7%, the verb 0.9%, and every content word below 7%. Native is exact in every slot. Slots follow the nine-word majority template; 10–11-word sentences shift the last positions.

7.1 EVERY INPUT, ONE OF NINE OUTPUTS

The grid below shows the 120 held-out sentences as individual squares. On the left, colour identifies which sentence the baseline emitted; the nine colours are the full output repertoire. On the right, every square is a different, exact sentence.



Each square is one test sentence in file order, 20 per row. Left colours group identical outputs; the largest group covers 36 inputs (30%). Right: every output is unique and correct, so the grid carries no clustering. Full prototype text in Appendix A.

> WHY THIS IS NOT “THE BASELINE NEEDS MORE TRAINING”
>
> The collapse persists at the end of training, when training loss is 1.32 and falling slowly. More steps would lower the loss by fitting the nine-sentence prior more tightly; they would not add input dependence. The information needed to distinguish the inputs is still spread across three times as many tokens, with no print-aligned starting point. The 256-token control (§9.1) pushes the same prior to 15 outputs and still misses all 120 sentences.

08 Error analysis

SIX REPRESENTATIVE ROWS, THEN MECHANISMS

TABLE 8 — SIDE-BY-SIDE EXAMPLES

#	REFERENCE	BASELINE OUTPUT	NATIVE OUTPUT
02	A dancer watched a bright lantern along the coast.	The fisherman closed the stone bridge before the storm.	A dancer watched a bright lantern along the coast.
04	A student watched the empty basket beside the harbour.	The fisherman carried a silver coin during the festival.	A student watched the empty basket beside the harbour.
09	The teacher finished ripe red apples under the old tree.	The captain recorded a silver coin in the morning.	The teacher finished ripe red apples under the old tree.
12	The artist measured warm woollen blankets after the rain.	The captain recorded a silver coin in the morning.	The artist measured warm woollen blankets after the rain.
37	The captain followed the tall ladder in the morning.	The fisherman closed the stone bridge before the storm.	The captain followed the tall ladder in the morning.
55	The tailor repaired several old books throughout the winter.	The captain recorded a silver coin in the morning.	The tailor repaired several old books throughout the winter.

Baseline is wrong on all 120 rows; these show the typical shape. Full table in `comparison.html` and `comparison.md`.

8.1 THREE MECHANISMS

Input bypass. The baseline emits one of nine sentences regardless of the dots. All 120 outputs start with “The”, although the references contain 87 “The”, 30 “A”, and 3 “My”. Its output vocabulary is 22 words; 116 of the 138 reference words never appear. The model has learned the corpus prior and stopped there.

Template completion. Position accuracy (Fig. 2) shows what it did learn: the final article 100%, the first article 70.5%, the second article 39.3%. These are the most predictable slots in a templated corpus. Content slots sit at 0.9–6.2%. It is completing a frame with frequent fragments, not decoding cells.

Vocabulary starvation. The nine prototypes are not verbatim training sentences — none of the nine appears in the training file. They are novel recombinations of high-frequency fragments: “fisherman”, “captain”, “stone bridge”, “silver coin”, “the storm”, “the morning”. That rules out simple memorisation of a single row while confirming that the learned object is a distribution over phrases, not a mapping from input.

8.2 IT IS NOT A NEAR MISS

The best baseline row matches 66.7% of its reference words and still has 30.8% character error; the median row is 64.0% CER. Bigram precision is 3.15%, so almost no two-word sequence survives. A model that was “nearly there” would show high unigram and moderate bigram overlap with input-sensitive variation. This model shows neither.

8.3 WHAT NATIVE’S ZERO ERROR DOES AND DOES NOT SHOW

Native is exact on all 120 rows, with 120 distinct outputs. That is strong evidence that the representation carries enough information to learn the mapping and that the model uses it: output diversity rules out a decoded prior. It is not evidence of open-domain translation. The test words are all in training, the patterns are the training patterns, and training loss of 0.002 is consistent with a near-deterministic transliteration of a closed distribution. The next test must introduce words and structures the model has not seen.

> THE DEFENSIBLE CLAIM
>
> On a closed, templated corpus, a 135M model with braille-native tokens and print-aligned embeddings learns braille→print exactly, while the same model with a stock tokenizer does not learn to condition on the input at all.

09 Validity and limitations

EVERY THREAT WE KNOW, AND ITS DIRECTION

Two results are robust: native is exact on this test set, and baseline does not condition on its input. The size of the gap is real. What the gap means beyond this corpus is constrained by the following.

TABLE 9 — THREATS TO VALIDITY

THREAT	QUANTIFICATION	DIRECTION	MITIGATION
Closed, templated corpus	138 words; 3 patterns; 100% of test words seen in training	Limits generalisation claim; does not affect the within-distribution comparison	Word- and pattern-disjoint test set; new domains
Baseline prompt truncation	292 / 480 train rows over 192 tokens; 2,485 tokens trimmed. Control at 256 tokens: 0 / 120 exact, CER 60.2%	Handicapped the baseline, but is not the cause	**Resolved.** 256-token control run; collapse persists
Token-budget asymmetry	max_len 128 vs 192	Favours baseline (more context), not native	Keep; report both budgets; same-budget test would truncate baseline more
Single seed, single recipe	1 run per arm; lr 1e-3 chosen after a failed 1e-4 run	Unknown variance; possible favourable seed	3 seeds; small lr sweep; report mean and interval
Small test set	n = 120; 95% Wilson interval for 120/120 is 96.9–100%; for 0/120 it is 0–3.1%	Adequate to separate the arms; too small to estimate rare errors	Larger held-out set; stratify by pattern
Greedy decoding only	No sampling or beam search	Unknown; likely small at this scale	Sampling stress test; report variance
Metrics	BLEU on 120 short sentences is coarse; CER averages per sentence	Could hide local errors; exact match mitigates	Human spot-check; error taxonomy by word class
UEB engine	Built-in Grade 1; liblouis absent; no Grade 2	Restricts scope; conclusions may not transfer to contracted braille	liblouis tables; Grade 2 evaluation
Round-trip filter	600 / 600 kept, 0 dropped	Clean labels (positive); selects engine-friendly strings	Document; test unfiltered edge cases separately

9.1 THE TRUNCATION CONFOUND, PRECISELY

At `max_len = 192`, 61% of baseline training rows exceeded the window and lost prompt tokens from the left: 2,485 tokens in total, 8.5 per affected row — about 2.8 braille cells out of 55, usually from the beginning. That is a real handicap, so the control was run rather than argued: the same recipe at `max_len = 256`, where **no training row is truncated** (mean loss 2.317, versus 2.421). It scores **0 / 120 exact**, CER **60.2%**, BLEU **4.17**, with **15** distinct outputs from a 29-word vocabulary. Extra context made the prior slightly richer and the sentences slightly closer; it did not create input dependence. The confound is eliminated, and the conclusion survives it.

9.2 WHAT THE CONFIDENCE INTERVALS SAY

With 120 items, 120/120 has a 95% Wilson interval of **96.9–100%**; 0/120 has **0–3.1%**. The test set is large enough to establish that the arms are not close. It is not large enough to estimate a 1% error rate, which matters for a production claim. Nothing here supports a per-character reliability claim beyond “no errors in 120 sentences”.

> BOTTOM LINE ON VALIDITY
>
> The comparison is internally valid: same model, data, optimizer, and seed, with one intended variable. Its external validity is low by construction. The report should be read as a controlled demonstration that the representation works — not as a benchmark of braille translation.

10 Decision and next steps

PROCEED · GATE 1 OPEN

Decision: proceed to a scoped next phase. The representation result is strong and the engineering path is clear. The baseline's failure has now survived a fair context budget; the remaining gate is lexical independence. The next phase should test that before scaling the recipe.

TABLE 10 — PRIORITISED PLAN

PRI	ACTION	WHY IT MATTERS	GATE / SUCCESS CRITERION
P0	Build an independent evaluation set: new words, new sentence structures, Grade 2, digits, punctuation, line breaks; split by word or pattern	Removes the largest threat: template and vocabulary overlap	Native must retain high exact match on unseen words/patterns; otherwise the result is memorisation
Done	Baseline control at `max_len 256`, same recipe	Tests the 61% truncation confound	**Passed:** 0 / 120 exact, CER 60.2% — the gap is not truncation
P1	Three seeds per arm; report mean and interval	Single-run results can flatter a recipe	Native exact match stable across seeds
P1	Scale to a 0.5B model on GPU (QLoRA if needed)	The user's intended model class; tests whether the representation advantage persists with capacity	Native ≥ baseline on the independent set at 0.5B
P1	Quantise the merged model (int8 / GGUF) and measure on-device latency	652 MB fp32 is not a deployable footprint for many readers	≤ 4× latency of fp32 at ≤ 1/3 size, no exact-match loss
P2	Tokenizer sweep: 150 vs 300–500 merges, unigram vs BPE	0.535 tokens/cell is good but not optimal; over-merging can hurt	Lower tokens/cell at equal exact match
P2	Human spot-check of 100 outputs; ambiguity and error taxonomy	Automatic metrics cannot judge acceptable variants	Documented agreement rate

10.1 ON EXPANDING THE CORPUS

Do not expand the current generator to 1,500–2,000 sentences as the next experiment. More combinations of the same 138 words would reduce training loss and make the model more confident on a distribution it already solves, while adding no evidence about reading. If the goal is a larger corpus, change its composition first: new lexical domains, varied syntax, verb tense and agreement, names, numbers, punctuation, and multi-line layout. Hold out words and patterns, not random rows. A 600-sentence diverse set is worth more than a 2,000-sentence templated one.

10.2 DEPLOYMENT NOTES

The native model ships as a merged fp32 checkpoint: 163.0M parameters, 652 MB, with an untied LM head created by the merge. That is larger than the 135M base (538 MB) because the head is duplicated. The adapter alternative is 1.86 MB on top of the base, but the native adapter also saves the resized embedding and head, so the merged artifact is the simpler delivery. For production, quantisation is the first lever; the 5.6× token reduction is the second, and it applies to every input regardless of model size.

> GATE STATUS, IN ONE SENTENCE
> Gate 2 is met: the 256-token baseline still scores 0 / 120. Gate 1 remains open: native must hold up on a word- or pattern-disjoint set. Until it does, present the result as a proof of representation, not as a translation benchmark.

11 Reproducing this run

ARTIFACTS · COMMANDS · CHECKS

TABLE 11 — DELIVERED ARTIFACTS

PATH	SIZE	CONTENTS
braillelm_native/	652,152,320 B	merged fp32 model, 163.0M params, tokenizer with 49,329 entries
out_baseline_adapter/	1,858,776 B	PEFT adapter weights plus tokenizer and config; needs the SmolLM2-135M base
data/	480 + 120	train.jsonl and test.jsonl, braille/print pairs
preds_baseline.tsv	120 rows	braille ⇥ predicted print
preds_native.tsv	120 rows	braille ⇥ predicted print
preds_baseline_256.tsv	120 rows	256-token control predictions, used in §9.1
comparison.html / .md	120 rows	full side-by-side table with filters
translate.py / translate.bat	–	run either checkpoint on new braille
braillelm_native_model.zip	661,311,094 B	portable archive of the merged model

COMMANDS

```
# translate new braille with the delivered merged model
translate.bat --text "⠓⠑⠇⠇⠕"
translate.bat --input my_braille.txt --out preds.tsv

# score on the held-out set
python translate.py --ckpt braillelm_native     --eval data/test.jsonl --cpu
python translate.py --ckpt out_baseline_adapter --eval data/test.jsonl --cpu

# retrain (from the project directory, HF cache warm)
python train.py --mode native   --model HuggingFaceTB/SmolLM2-135M \
  --data data --cpu --epochs 1 --lr 1e-3 --batch 4 --max_len 128
python train.py --mode baseline --model HuggingFaceTB/SmolLM2-135M \
  --data data --cpu --epochs 1 --lr 1e-3 --batch 4 --max_len 192
```

The adapter requires the base model weights; the merged checkpoint is standalone. Set `PYTHONIOENCODING=utf-8` on Windows consoles before printing braille.

> WHAT REPRODUCING THIS ESTABLISHES
>
> A third party can rescore both delivered checkpoints against the same 120 sentences and recover every number in §6–§7. The commands above do not retrain; they evaluate the artifacts exactly as shipped. Retraining requires the Hugging Face cache and takes roughly 12 minutes on this hardware.

VERIFICATION CHECKLIST

CHECK	HOW	EXPECTED
Split sizes	wc -l data/train.jsonl data/test.jsonl	480 · 120
Prediction rows	wc -l preds_native.tsv preds_baseline.tsv	120 · 120
Native exact match	translate.py --ckpt braillelm_native --eval …	120 / 120
Baseline BLEU · CER	translate.py --ckpt out_baseline_adapter --eval …	4.0 · 0.638

Not included in the delivery: training logs, the Hugging Face cache, and any GPU run. The numbers in this report come from the CPU checkpoints listed in Table 11.

12 Model and artifacts

WHAT WAS TRAINED · WHAT SHIPS

The delivered native model is a merged SmolLM2-135M checkpoint with a braille-extended vocabulary. The baseline is a PEFT adapter on the stock tokenizer. Both are fp32 CPU models; neither needs a GPU to run, and both can be evaluated from the files listed here.

TABLE 12 — MODEL FACTS

Base checkpoint	HuggingFaceTB/SmolLM2-135M
Architecture	LlamaForCausalLM, decoder-only
Layers · hidden · FFN	30 · 576 · 1,536
Attention heads · KV	9 · 3 (GQA)
Context window	8,192 tokens (trained at 128)
Vocabulary	49,152 → 49,329 (+177)
Merged parameters	163,030,464
Merged size, fp32	652,152,320 B
Output head	untied after merge
Adapter	LoRA r8 · α16 · q_proj, v_proj
Saved modules	embed_tokens, lm_head
Precision	fp32, CPU
Decoding	greedy, EOS = 0
Base licence	Apache-2.0

TABLE 13 — SHIPPED ARTIFACTS

PATH	SIZE	USE
braillelm_native/	652.2 MB	standalone merged model
out_baseline_adapter/	1.86 MB	baseline adapter; needs base
data/	600 pairs	train and test splits
preds_native.tsv	120 rows	native predictions
preds_baseline.tsv	120 rows	baseline predictions
preds_baseline_256.tsv	120 rows	control predictions
llm_comparison/	11 files	test set, prompt, scorer, brief
comparison_pack.json	71 KB	all rows and scores in one file
translate.py	5.7 KB	run or evaluate a checkpoint

12.1 INTENDED USE, STATED NARROWLY

Intended. Short, uncontracted UEB Grade 1 sentences drawn from the 138-word templated family used in training; CPU inference at batch 1–8; comparing tokenisation and embedding strategies at small model scale.

Not intended. Grade 2 contractions, digits, punctuation beyond the period, line breaks, multi-sentence documents, literary or technical braille, or production accessibility use. Unseen words and syntax are untested (§9, §10).

12.2 DESIGN CHOICES AND WHAT THEY COST

CHOICE	WHAT IT BUYS	WHAT IT COSTS
177 braille tokens	0.535 tokens per cell instead of 3.00; prompts fit 128 tokens	+177 embedding rows; a merged checkpoint larger than the base
BKFT initialisation	print-aligned starting rows; convergence below 0.3 by step 30	depends on a print expansion for every added token
LoRA on q_proj, v_proj	460,800 baseline parameters; fast CPU training	limits capacity; not a full fine-tune comparison
Save embeddings and head	new rows actually learn	PEFT reports 29.85% “trainable”; the artifact is 163M params
Left truncation at max_len	answer tokens never cut	baseline lost prompt tokens in 61% of rows; tested by the 256 control

12.3 TRAINING LINEAGE

RUN	MAX_LEN	STEPS	EPOCH MEAN LOSS	FINAL STEP	WALL CLOCK
Baseline, delivered	192	120	2.421	1.321	≈ 8 min
Native, delivered	128	120	0.501	0.002	≈ 4 min
Baseline, 256 control	256	120	2.317	–	–

All runs: seed 0, AdamW at lr 1e-3, batch 4, one epoch, fp32 CPU.

> MODEL SIZE, PRECISELY
>
> The base is a 135M-parameter model. The merged native artifact is **163,030,464 parameters** because the merge unties the output head and materialises a full copy: 652 MB versus 538 MB for the base. The adapter form is 1.86 MB plus the base. Quote the merged number when discussing footprint, and the base number when discussing capacity.

13 Data card and vocabulary

600 SYNTHETIC SENTENCES · CLOSED CLASS

The corpus is fully specified and synthetic: every sentence comes from a fixed bank of lexical classes and one of three templates. It is a controlled probe of representation, not a sample of braille in the world.

TABLE 14 — DATA CARD

Source	sentences.txt → generated list
Size	600 unique sentences
Split	480 train / 120 test, seed 0
Vocabulary	138 unique words, closed
Sentence shapes	3 · 525 / 72 / 3 by length
Length	9.13 words · 54.27 chars
Braille length	55.27 cells, UEB Grade 1
Character set	A–Z, space, “.”
Round trip	600 / 600 kept, 0 dropped
Split overlap	0 braille · 0 print strings
Test coverage	138 / 138 words seen in train
Privacy	synthetic; no personal data

TABLE 15 — LEXICAL CLASSES

CLASS	N	EXAMPLES
Actors	28	artist, baker, captain, dancer
Verbs	30	arranged, carried, closed
Modifiers	23	bright, empty, fresh, heavy
Objects	20	apples, basket, blankets
Places	15	barn, bridge, coast, garden
Prepositions	15	above, across, after, along
Determiners	3	a, my, the
Quantifiers	3	many, several, some
Person nouns	2	girl, man

Memberships sum to 139 because “bridge” is both an object and a place; the unique vocabulary is 138 words.

13.1 FULL VOCABULARY, WITH TRAINING FREQUENCY

a	322	child	17	gathered	14	nurse	21	started	19
above	33	clock	27	girl	17	old	87	stone	33
across	40	closed	25	glass	32	opened	14	storm	48
after	48	coast	42	green	38	painted	23	student	16
along	42	coin	27	harbour	33	painter	24	studied	18
apples	35	collected	24	heavy	28	paper	24	tailor	26
around	32	counted	16	hunter	21	pilot	17	tall	30
arranged	19	dancer	29	in	41	planted	21	teacher	14
artist	25	delivered	19	inside	37	polished	22	the	1301
baker	14	described	19	iron	26	prepared	18	throughout	38
barn	31	discovered	30	ladder	30	protected	19	traveller	27
basket	31	doctor	23	lantern	27	quiet	30	tree	44
before	48	during	39	librarian	17	rain	48	umbrella	32
behind	31	empty	31	man	17	recorded	25	under	44
beneath	44	engineer	15	many	33	red	35	valley	40
beside	33	examined	21	map	24	remembered	19	village	33
blankets	25	explained	17	market	32	repaired	19	warm	25
boat	27	farmer	17	measured	20	ripe	35	watched	18
books	26	festival	39	mentioned	23	river	50	weaver	20
box	28	field	38	merchant	19	road	32	wet	32
bread	37	finished	16	morning	41	sailor	16	window	32
bridge	77	fisherman	20	museum	37	scientist	20	winter	38
bright	27	folded	9	musician	21	several	26	wooden	33
broken	27	followed	24	my	21	shopkeeper	18	woollen	25
captain	30	fresh	37	narrow	32	silver	27	writer	16
carpenter	23	garden	30	near	50	sketched	23	young	17
carried	23	gardener	19	neighbour	21	small	27		
chairs	33	gate	26	noticed	23	some	37		

Counts are occurrences across the 480 training sentences. The test split uses the same 138 words, so no evaluation word is out of vocabulary. Sorted alphabetically; word and count.

Held-out set · 1 / 5

Every held-out item, in file order. **B** and **C** are its baseline and control prototype ranks; **CER** is the baseline's per-row error. Prototypes: Appendix A and §20.3. Fields: `test_set.jsonl`.

TABLE 16 — TEST ITEMS 001–024

#	BRAILLE INPUT / ENGLISH REFERENCE	B	C	CER
001	⠠⠙⠑ ⠍⠁⠗⠙⠑⠝⠑⠗ ⠎⠞⠁⠗⠞⠑⠙ ⠁ ⠺⠑⠞ ⠥⠍⠃⠗⠑⠇⠇⠁ ⠃⠑⠋⠕⠗⠑ ⠞⠓⠑ ⠎⠞⠕⠗⠍⠲	3	1	49.1%
	The gardener started a wet umbrella before the storm.			
002	⠠ ⠙⠁⠝⠉⠑⠗ ⠺⠁⠞⠉⠓⠑⠙ ⠁ ⠃⠗⠊⠛⠓⠞ ⠇⠁⠝⠞⠑⠗⠝ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	1	3	80.0%
	A dancer watched a bright lantern along the coast.			
003	⠠⠞⠓⠑ ⠇⠊⠃⠗⠁⠗⠊⠁⠝ ⠗⠑⠉⠕⠗⠙⠑⠙

TABLE 17 — TEST ITEMS 025–048

#	BRAILLE INPUT / ENGLISH REFERENCE	B	C	CER
025	⠠⠓⠑ ⠋⠁⠗⠍⠑⠗ ⠍⠑⠝⠞⠊⠕⠝⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	3	2	63.0%
	The farmer mentioned some fresh bread along the coast.			
026	⠠⠓⠑ ⠙⠕⠉⠞⠕⠗ ⠁⠗⠗⠁⠝⠛⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	3	1	68.5%
	The doctor arranged several old books along the coast.			
027	⠠⠓⠑ ⠓⠥⠝⠞⠑⠗ ⠁⠗⠗⠁⠝⠛⠑⠙ ⠍⠁⠝⠽ ⠺⠕⠕⠙⠑⠝ ⠉⠓⠁⠊⠗⠎ ⠙⠥⠗⠊⠝⠛ ⠞⠓⠑ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	2	7	59.3%
	The hunter arranged many wooden chairs during the festival.			
028	⠠⠁ ⠎⠉⠊⠑⠝⠞⠊⠎⠞ ⠏⠇⠁⠝⠞⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	3	3	75.5%
	A scientist planted several old books near the river.			
029	⠠⠓⠑ ⠞⠁⠊⠇⠕⠗ ⠎⠞⠁⠗⠞⠑⠙ ⠞⠓⠑ ⠟⠥⠊⠑⠞ ⠛⠁⠗⠙⠑⠝ ⠙⠥⠗⠊⠝⠛ ⠞⠓⠑ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	5	1	48.2%
	The tailor started the quiet garden during the festival.			
030	⠠⠁ ⠎⠓⠕⠏⠅⠑⠑⠏⠑⠗ ⠏⠕⠇⠊⠎⠓⠑⠙ ⠞⠓⠑ ⠑⠍⠏⠞⠽ ⠃⠁⠎⠅⠑⠞ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	3	12	66.7%
	A shopkeeper polished the empty basket near the river.			
031	⠠⠓⠑ ⠝⠥⠗⠎⠑ ⠍⠑⠁⠎⠥⠗⠑⠙ ⠞⠓⠑ ⠟⠥⠊⠑⠞ ⠛⠁⠗⠙⠑⠝ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	4	1	60.8%
	The nurse measured the quiet garden near the river.			
032	⠠⠓⠑ ⠞⠑⠁⠉⠓⠑⠗ ⠗⠑⠏⠁⠊⠗⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	4	1	64.2%
	The teacher repaired some fresh bread near the river.			
033	⠠⠁ ⠎⠉⠊⠑⠝⠞⠊⠎⠞ ⠗⠑⠍⠑⠍⠃⠑⠗⠑⠙ ⠞⠓⠑ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠃⠑⠓⠊⠝⠙ ⠞⠓⠑ ⠃⠁⠗⠝⠲	1	3	63.0%
	A scientist remembered the small boat behind the barn.			
034	⠠⠓⠑ ⠙⠕⠉⠞⠕⠗ ⠋⠊⠝⠊⠎⠓⠑⠙ ⠞⠓⠑ ⠛⠗⠑⠑⠝ ⠋⠊⠑⠇⠙ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	4	1	58.8%
	The doctor finished the green field near the river.			
035	⠠⠁ ⠙⠁⠝⠉⠑⠗ ⠝⠕⠞⠊⠉⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠁⠉⠗⠕⠎⠎ ⠞⠓⠑ ⠧⠁⠇⠇⠑⠽⠲	3	3	77.4%
	A dancer noticed several old books across the valley.			
036	⠠⠓⠑ ⠺⠗⠊⠞⠑⠗ ⠏⠕⠇⠊⠎⠓⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲	7	2	63.6%
	The writer polished some fresh bread inside the museum.			
037	⠠⠓⠑ ⠉⠁⠏⠞⠁⠊⠝ ⠋⠕⠇⠇⠕⠺⠑⠙ ⠞⠓⠑ ⠇⠁⠙⠙⠑⠗ ⠊⠝ ⠞⠓⠑ ⠍⠕⠗⠝⠊⠝⠛⠲	1	1	63.5%
	The captain followed the tall ladder in the morning.			
038	⠠⠓⠑ ⠛⠁⠗⠙⠑⠝⠑⠗ ⠁⠗⠗⠁⠝⠛⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲	2	2	63.2%
	The gardener arranged some fresh bread inside the museum.			
039	⠠⠓⠑ ⠺⠗⠊⠞⠑⠗ ⠏⠗⠕⠞⠑⠉⠞⠑⠙ ⠞⠓⠑ ⠎⠞⠕⠝⠑ ⠃⠗⠊⠙⠛⠑ ⠞⠓⠗⠕⠥⠛⠓⠕⠥⠞ ⠞⠓⠑ ⠺⠊⠝⠞⠑⠗⠲	2	6	65.0%
	The writer protected the stone bridge throughout the winter.			
040	⠠⠁ ⠏⠁⠊⠝⠞⠑⠗ ⠋⠊⠝⠊⠎⠓⠑⠙ ⠁ ⠎⠊⠇⠧⠑⠗ ⠉⠕⠊⠝ ⠁⠋⠞⠑⠗ ⠞⠓⠑ ⠗⠁⠊⠝⠲	1	9	77.1%
	A painter finished a silver coin after the rain.			
041	⠠⠁ ⠙⠁⠝⠉⠑⠗ ⠎⠞⠁⠗⠞⠑⠙ ⠞⠓⠑ ⠊⠗⠕⠝ ⠛⠁⠞⠑ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲	6	3	73.5%
	A dancer started the iron gate inside the museum.			
042	⠠⠓⠑ ⠓⠥⠝⠞⠑⠗ ⠕⠏⠑⠝⠑⠙ ⠗⠊⠏⠑ ⠗⠑⠙ ⠁⠏⠏⠇⠑⠎ ⠁⠃⠕⠧⠑ ⠞⠓⠑ ⠧⠊⠇⠇⠁⠛⠑⠲	1	1	69.2%
	The hunter opened ripe red apples above the village.			
043	⠠⠓⠑ ⠞⠗⠁⠧⠑⠇⠇⠑⠗ ⠍⠑⠁⠎⠥⠗⠑⠙ ⠞⠓⠑ ⠑⠍⠏⠞⠽ ⠃⠁⠎⠅⠑⠞ ⠁⠃⠕⠧⠑ ⠞⠓⠑ ⠧⠊⠇⠇⠁⠛⠑⠲	2	1	69.0%
	The traveller measured the empty basket above the village.			
044	⠠⠁ ⠎⠁⠊⠇⠕⠗ ⠉⠁⠗⠗⠊⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠃⠑⠋⠕⠗⠑ ⠞⠓⠑ ⠎⠞⠕⠗⠍⠲	1	8	52.9%
	A sailor carried some fresh bread before the storm.			
045	⠠⠁ ⠏⠁⠊⠝⠞⠑⠗ ⠋⠕⠇⠇⠕⠺⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	1	8	66.7%
	A painter followed some fresh bread near the river.			
046	⠠⠓⠑ ⠋⠊⠎⠓⠑⠗⠍⠁⠝ ⠋⠊⠝⠊⠎⠓⠑⠙ ⠞⠓⠑ ⠇⠁⠙⠙⠑⠗ ⠁⠉⠗⠕⠎⠎ ⠞⠓⠑ ⠧⠁⠇⠇⠑⠽⠲	2	2	68.4%
	The fisherman finished the tall ladder across the valley.			
047	⠠⠓⠑ ⠍⠑⠗⠉⠓⠁⠝⠞ ⠗⠑⠏⠁⠊⠗⠑⠙ ⠞⠓⠑ ⠛⠇⠁⠎⠎ ⠺⠊⠝⠙⠕⠺ ⠃⠑⠋⠕⠗⠑ ⠞⠓⠑ ⠎⠞⠕⠗⠍⠲	5	2	41.1%
	The merchant repaired the glass window before the storm.			
048	⠠⠁ ⠉⠓⠊⠇⠙ ⠑⠭⠏⠇⠁⠊⠝⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠙⠥⠗⠊⠝⠛ ⠞⠓⠑ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	1	13	75.0%
	A child explained several old books during the festival.			

B / C are the baseline and control prototype ranks; rows 025–048 continue the same file order.

TABLE 18 — TEST ITEMS 049–072

#	BRAILLE INPUT / ENGLISH REFERENCE	B	C	CER
049	⠠⠞⠓⠑ ⠇⠊⠃⠗⠁⠗⠊⠁⠝ ⠎⠞⠁⠗⠞⠑⠙ ⠞⠓⠑ ⠃⠗⠕⠅⠑⠝ ⠉⠇⠕⠉⠅ ⠃⠑⠋⠕⠗⠑ ⠞⠓⠑ ⠎⠞⠕⠗⠍⠲	2	1	60.7%
	The librarian started the broken clock before the storm.			
050	⠠⠞⠓⠑ ⠑⠝⠛⠊⠝⠑⠑⠗ ⠏⠕⠇⠊⠎⠓⠑⠙ ⠁ ⠺⠑⠞ ⠥⠍⠃⠗⠑⠇⠇⠁ ⠥⠝⠙⠑⠗ ⠞⠓⠑ ⠕⠇⠙ ⠞⠗⠑⠑⠲	1	1	66.1%
	The engineer polished a wet umbrella under the old tree.			
051	⠠⠞⠓⠑ ⠇⠊⠃⠗⠁⠗⠊⠁⠝ ⠁⠗⠗⠁⠝⠛⠑⠙ ⠞⠓⠑ ⠛⠇⠁⠎⠎ ⠺⠊⠝⠙⠕⠺ ⠊⠝ ⠞⠓⠑ ⠍⠕⠗⠝⠊⠝⠛⠲	7	2	43.6%
	The librarian arranged the glass window in the morning.			
052	⠠⠞⠓⠑ ⠍⠑⠗⠉⠓⠁⠝⠞ ⠉⠕⠥⠝⠞⠑⠙ ⠞⠓⠑ ⠑⠍⠏⠞⠽ ⠃⠁⠎⠅⠑⠞ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	4	1	58.5%
	The merchant counted the empty basket near the river.			
053	⠠⠞⠓⠑ ⠽⠕⠥⠝⠛ ⠛⠊⠗⠇ ⠗⠑⠍⠑⠍⠃⠑⠗⠑⠙ ⠁ ⠺⠑⠞ ⠥⠍⠃⠗⠑⠇⠇⠁ ⠁⠉⠗⠕⠎⠎ ⠞⠓⠑ ⠧⠁⠇⠇⠑⠽⠲	2	7	67.8%
	The young girl remembered a wet umbrella across the valley.			
054	⠠⠞⠓⠑ ⠓⠥⠝⠞⠑⠗ ⠍⠑⠝⠞⠊⠕⠝⠑⠙ ⠞⠓⠑ ⠺⠊⠝⠙⠕⠺ ⠁⠃⠕⠧⠑ ⠞⠓⠑ ⠃⠗⠊⠙⠛⠑⠲	2	2	63.2%
	The hunter mentioned the glass window beneath the bridge.			
055	⠠⠞⠓⠑ ⠞⠁⠊⠇⠕⠗ ⠗⠑⠏⠁⠊⠗⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠞⠓⠗⠕⠥⠛⠓⠕⠥⠞ ⠞⠓⠑ ⠺⠊⠝⠞⠑⠗⠲	2	1	66.7%
	The tailor repaired several old books throughout the winter.			
056	⠠⠞⠓⠑ ⠞⠁⠊⠇⠕⠗ ⠉⠇⠕⠎⠑⠙ ⠞⠓⠑ ⠎⠞⠕⠝⠑ ⠃⠗⠊⠙⠛⠑ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	4	1	58.8%
	The tailor closed the stone bridge along the coast.			
057	⠠ ⠙⠁⠝⠉⠑⠗ ⠺⠁⠞⠉⠓⠑⠙ ⠞⠓⠑ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠁⠉⠗⠕⠎⠎ ⠞⠓⠑ ⠧⠁⠇⠇⠑⠽⠲	1	3	68.0%
	A dancer watched the small boat across the valley.			
058	⠠⠞⠓⠑ ⠍⠑⠗⠉⠓⠁⠝⠞ ⠋⠕⠇⠇⠕⠺⠑⠙ ⠞⠓⠑ ⠝⠁⠗⠗⠕⠺ ⠗⠕⠁⠙ ⠃⠑⠎⠊⠙⠑ ⠞⠓⠑ ⠓⠁⠗⠃⠕⠥⠗⠲	2	1	63.2%
	The merchant followed the narrow road beside the harbour.			
059	⠠ ⠙⠁⠝⠉⠑⠗ ⠋⠕⠇⠙⠑⠙ ⠞⠓⠑ ⠺⠊⠝⠙⠕⠺ ⠃⠑⠓⠊⠝⠙ ⠞⠓⠑ ⠃⠁⠗⠝⠲	6	3	44.9%
	A dancer folded the glass window behind the barn.			
060	⠠⠞⠓⠑ ⠇⠊⠃⠗⠁⠗⠊⠁⠝ ⠝⠕⠞⠊⠉⠑⠙ ⠞⠓⠑ ⠟⠥⠊⠑⠞ ⠛⠁⠗⠙⠑⠝ ⠞⠓⠗⠕⠥⠛⠓⠕⠥⠞ ⠞⠓⠑ ⠺⠊⠝⠞⠑⠗⠲	2	5	65.6%
	The librarian noticed the quiet garden throughout the winter.			
061	⠠⠞⠓⠑ ⠞⠗⠁⠧⠑⠇⠇⠑⠗ ⠑⠭⠁⠍⠊⠝⠑⠙ ⠗⠊⠏⠑ ⠗⠑⠙ ⠁⠏⠏⠇⠑⠎ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	3	1	68.5%
	The traveller examined ripe red apples near the river.			
062	⠠⠞⠓⠑ ⠋⠁⠗⠍⠑⠗ ⠙⠊⠎⠉⠕⠧⠑⠗⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠁⠗⠕⠥⠝⠙ ⠞⠓⠑ ⠍⠁⠗⠅⠑⠞⠲	2	1	62.1%
	The farmer discovered several old books around the market.			
063	⠠⠞⠓⠑ ⠉⠁⠏⠞⠁⠊⠝ ⠕⠏⠑⠝⠑⠙ ⠁ ⠏⠁⠏⠑⠗ ⠍⠁⠏ ⠁⠃⠕⠧⠑ ⠞⠓⠑ ⠧⠊⠇⠇⠁⠛⠑⠲	6	1	75.5%
	The captain opened a paper map above the village.			
064	⠠ ⠎⠁⠊⠇⠕⠗ ⠙⠑⠎⠉⠗⠊⠃⠑⠙ ⠍⠁⠝⠽ ⠺⠕⠕⠙⠑⠝ ⠉⠓⠁⠊⠗⠎ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	1	3	72.2%
	A sailor described many wooden chairs along the coast.			
065	⠠⠞⠓⠑ ⠁⠗⠞⠊⠎⠞ ⠗⠑⠏⠁⠊⠗⠑⠙ ⠗⠊⠏⠑ ⠗⠑⠙ ⠁⠏⠏⠇⠑⠎ ⠁⠃⠕⠧⠑ ⠞⠓⠑ ⠧⠊⠇⠇⠁⠛⠑⠲	3	2	66.7%
	The artist repaired ripe red apples above the village.			
066	⠠⠞⠓⠑ ⠛⠁⠗⠙⠑⠝⠑⠗ ⠉⠕⠥⠝⠞⠑⠙ ⠞⠓⠑ ⠝⠁⠗⠗⠕⠺ ⠗⠕⠁⠙ ⠁⠉⠗⠕⠎⠎ ⠞⠓⠑ ⠧⠁⠇⠇⠑⠽⠲	7	2	69.1%
	The gardener counted the narrow road across the valley.			
067	⠠ ⠎⠓⠕⠏⠅⠑⠑⠏⠑⠗ ⠏⠗⠕⠞⠑⠉⠞⠑⠙ ⠞⠓⠑ ⠎⠞⠕⠝⠑ ⠃⠗⠊⠙⠛⠑ ⠁⠃⠕⠧⠑ ⠞⠓⠑ ⠧⠊⠇⠇⠁⠛⠑⠲	2	4	72.4%
	A shopkeeper protected the stone bridge above the village.			
068	⠠⠞⠓⠑ ⠞⠑⠁⠉⠓⠑⠗ ⠙⠑⠎⠉⠗⠊⠃⠑⠙ ⠞⠓⠑ ⠝⠁⠗⠗⠕⠺ ⠗⠕⠁⠙ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲	1	2	57.1%
	The teacher described the narrow road inside the museum.			
069	⠠⠞⠓⠑ ⠞⠗⠁⠧⠑⠇⠇⠑⠗ ⠁⠗⠗⠁⠝⠛⠑⠙ ⠁ ⠓⠑⠁⠧⠽ ⠃⠕⠭ ⠕⠝ ⠞⠓⠑ ⠍⠕⠗⠝⠊⠝⠛⠲	1	1	78.0%
	The traveller arranged a heavy box in the morning.			
070	⠠⠞⠓⠑ ⠉⠁⠏⠞⠁⠊⠝ ⠎⠞⠥⠙⠊⠑⠙ ⠞⠓⠑ ⠛⠗⠑⠑⠝ ⠋⠊⠑⠇⠙ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲	4	1	45.1%
	The captain studied the green field near the river.			
071	⠠ ⠎⠞⠥⠙⠑⠝⠞ ⠎⠞⠁⠗⠞⠑⠙ ⠞⠓⠑ ⠑⠍⠏⠞⠽ ⠃⠁⠎⠅⠑⠞ ⠃⠑⠝⠑⠁⠞⠓ ⠞⠓⠑ ⠃⠗⠊⠙⠛⠑⠲	9	4	66.7%
	A student started the empty basket beneath the bridge.			
072	⠠⠞⠓⠑ ⠞⠗⠁⠧⠑⠇⠇⠑⠗ ⠙⠑⠇⠊⠧⠑⠗⠑⠙ ⠞⠓⠑ ⠃⠗⠕⠅⠑⠝ ⠉⠇⠕⠉⠅ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲	2	1	59.3%
	The traveller delivered the broken clock inside the museum.			

The 10- and 11-word sentences cluster in the later items; slot-2 accuracy in §7 is computed only on the 112 nine-word rows.

TABLE 19 — TEST ITEMS 073–096

#	BRAILLE INPUT / ENGLISH REFERENCE	B	C	CER
073	⠠⠮ ⠏⠊⠇⠕⠞ ⠎⠞⠥⠙⠊⠑⠙ ⠁ ⠃⠗⠊⠛⠓⠞ ⠇⠁⠝⠞⠑⠗⠝ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	4	1	70.6%
	The pilot studied a bright lantern along the coast.			
074	⠠⠮ ⠏⠊⠇⠕⠞ ⠗⠑⠏⠁⠊⠗⠑⠙ ⠮ ⠟⠥⠊⠑⠞ ⠛⠁⠗⠙⠑⠝ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	1	1	63.5%
	The pilot repaired the quiet garden along the coast.			
075	⠠ ⠉⠁⠗⠏⠑⠝⠞⠑⠗ ⠺⠁⠞⠉⠓⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠙⠥⠗⠊⠝⠛ ⠮ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	2	4	70.2%
	A carpenter watched some fresh bread during the festival.			
076	⠠ ⠍⠥⠎⠊⠉⠊⠁⠝ ⠏⠗⠕⠞⠑⠉⠞⠑⠙ ⠮ ⠞⠁⠇⠇ ⠇⠁⠙⠙⠑⠗ ⠙⠥⠗⠊⠝⠛ ⠮ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	2	4	66.7%
	A musician protected the tall ladder during the festival.			
077	⠠⠮ ⠞⠗⠁⠧⠑⠇⠇⠑⠗ ⠕⠏⠑⠝⠑⠙ ⠮ ⠑⠍⠏⠞⠽ ⠃⠁⠎⠅⠑⠞ ⠁⠗⠕⠥⠝⠙ ⠮ ⠍⠁⠗⠅⠑⠞⠲	5	2	55.4%
	The traveller opened the empty basket around the market.			
078	⠠⠮ ⠇⠊⠃⠗⠁⠗⠊⠁⠝ ⠉⠁⠗⠗⠊⠑⠙ ⠁ ⠺⠁⠞ ⠥⠍⠃⠗⠑⠇⠇⠁ ⠊⠝⠎⠊⠙⠑ ⠮ ⠍⠥⠎⠑⠥⠍⠲	7	2	60.0%
	The librarian carried a wet umbrella inside the museum.			
079	⠠ ⠍⠥⠎⠊⠉⠊⠁⠝ ⠗⠑⠏⠁⠊⠗⠑⠙ ⠮ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠃⠑⠓⠊⠝⠙ ⠮ ⠃⠁⠗⠝⠲	1	8	62.7%
	A musician repaired the small boat behind the barn.			
080	⠠⠮ ⠃⠁⠅⠑⠗ ⠋⠕⠇⠇⠕⠺⠑⠙ ⠮ ⠟⠥⠊⠑⠞ ⠛⠁⠗⠙⠑⠝ ⠁⠇⠕⠝⠛ ⠞⠓⠑ ⠉⠕⠁⠎⠞⠲	1	1	57.7%
	The baker followed the quiet garden along the coast.			
081	⠠⠮ ⠙⠕⠉⠞⠕⠗ ⠑⠭⠁⠍⠊⠝⠑⠙ ⠮ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠁⠃⠕⠧⠑ ⠮ ⠧⠊⠇⠇⠁⠛⠑⠲	3	1	64.2%
	The doctor examined the small boat above the village.			
082	⠠ ⠋⠕⠇⠙⠑⠗ ⠏⠁⠊⠝⠞⠑⠙ ⠁ ⠓⠑⠁⠧⠽ ⠃⠕⠭ ⠁⠇⠕⠝⠛ ⠮ ⠍⠁⠗⠅⠑⠞⠲	1	14	82.2%
	A painter folded a heavy box along the coast.			
083	⠠⠮ ⠋⠁⠗⠍⠑⠗ ⠗⠑⠍⠑⠍⠃⠑⠗⠑⠙ ⠮ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠙⠥⠗⠊⠝⠛ ⠮ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	2	2	59.6%
	The farmer remembered the small boat during the festival.			
084	⠠⠮ ⠍⠑⠗⠉⠓⠁⠝⠞ ⠉⠁⠗⠗⠊⠑⠙ ⠮ ⠛⠇⠁⠎⠎ ⠺⠊⠝⠙⠕⠺ ⠁⠗⠕⠥⠝⠙ ⠮ ⠍⠁⠗⠅⠑⠞⠲	5	2	60.7%
	The merchant carried the glass window around the market.			
085	⠠⠮ ⠙⠕⠉⠞⠕⠗ ⠎⠅⠑⠞⠉⠓⠑⠙ ⠮ ⠎⠞⠕⠝⠑ ⠃⠗⠊⠙⠛⠑ ⠝⠑⠁⠗ ⠮ ⠗⠊⠧⠑⠗⠲	1	1	42.3%
	The doctor sketched the stone bridge near the river.			
086	⠠⠮ ⠋⠊⠎⠓⠑⠗⠍⠁⠝ ⠍⠑⠁⠎⠥⠗⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠁⠇⠕⠝⠛ ⠮ ⠉⠕⠁⠎⠞⠲	5	2	62.5%
	The fisherman measured some fresh bread along the coast.			
087	⠠⠮ ⠛⠁⠗⠙⠑⠝⠑⠗ ⠙⠊⠎⠉⠕⠧⠑⠗⠑⠙ ⠁ ⠺⠁⠞ ⠥⠍⠃⠗⠑⠇⠇⠁ ⠃⠑⠓⠊⠝⠙ ⠮ ⠃⠁⠗⠝⠲	3	2	65.5%
	The gardener discovered a wet umbrella behind the barn.			
088	⠠ ⠎⠉⠊⠑⠝⠞⠊⠎⠞ ⠝⠕⠞⠊⠉⠑⠙ ⠁ ⠓⠑⠁⠧⠽ ⠃⠕⠭ ⠃⠑⠓⠊⠝⠙ ⠮ ⠃⠁⠗⠝⠲	1	9	79.2%
	A scientist noticed a heavy box behind the barn.			
089	⠠⠮ ⠋⠊⠎⠓⠑⠗⠍⠁⠝ ⠕⠏⠑⠝⠑⠙ ⠮ ⠎⠞⠕⠝⠑ ⠃⠗⠊⠙⠛⠑ ⠞⠓⠗⠕⠥⠛⠓⠕⠥⠞ ⠮ ⠺⠊⠝⠞⠑⠗⠲	2	6	68.3%
	The fisherman opened the stone bridge throughout the winter.			
090	⠠⠮ ⠃⠁⠅⠑⠗ ⠙⠑⠎⠉⠗⠊⠃⠑⠙ ⠮ ⠃⠗⠕⠅⠑⠝ ⠉⠇⠕⠉⠅ ⠃⠑⠓⠊⠝⠙ ⠮ ⠃⠁⠗⠝⠲	4	1	60.4%
	The baker described the broken clock behind the barn.			
091	⠠⠮ ⠺⠑⠁⠧⠑⠗ ⠋⠕⠇⠇⠕⠺⠑⠙ ⠮ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠁⠉⠗⠕⠎⠎ ⠮ ⠧⠁⠇⠇⠑⠽⠲	4	1	62.3%
	The weaver followed the small boat across the valley.			
092	⠠⠮ ⠕⠇⠙ ⠍⠁⠝ ⠗⠑⠉⠕⠗⠙⠑⠙ ⠁ ⠃⠗⠊⠛⠓⠞ ⠇⠁⠝⠞⠑⠗⠝ ⠙⠥⠗⠊⠝⠛ ⠮ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	2	7	50.0%
	The old man recorded a bright lantern during the festival.			
093	⠠⠮ ⠉⠁⠏⠞⠁⠊⠝ ⠗⠑⠉⠕⠗⠙⠑⠙ ⠮ ⠟⠥⠊⠑⠞ ⠛⠁⠗⠙⠑⠝ ⠁⠉⠗⠕⠎⠎ ⠮ ⠧⠁⠇⠇⠑⠽⠲	5	2	53.6%
	The captain recorded the quiet garden across the valley.			
094	⠠⠮ ⠽⠕⠥⠝⠛ ⠛⠊⠗⠇ ⠎⠞⠁⠗⠞⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠙⠥⠗⠊⠝⠛ ⠮ ⠋⠑⠎⠞⠊⠧⠁⠇⠲	2	6	63.9%
	The young girl started several old books during the festival.			
095	⠠⠮ ⠍⠑⠗⠉⠓⠁⠝⠞ ⠑⠭⠁⠍⠊⠝⠑⠙ ⠺⠁⠗⠍ ⠺⠕⠕⠇⠇⠑⠝ ⠃⠇⠁⠝⠅⠑⠞⠎ ⠃⠑⠎⠊⠙⠑ ⠮ ⠓⠁⠗⠃⠕⠥⠗⠲	5	5	60.3%
	The merchant examined warm woollen blankets beside the harbour.			
096	⠠ ⠎⠁⠊⠇⠕⠗ ⠍⠑⠁⠎⠥⠗⠑⠙ ⠮ ⠎⠞⠕⠝⠑ ⠃⠗⠊⠙⠛⠑ ⠃⠑⠋⠕⠗⠑ ⠮ ⠎⠞⠕⠗⠍⠲	1	3	30.8%
	A sailor measured the stone bridge before the storm.			

Rows 073–096 continue the same order. Nothing in these pages appears in `data/train.jsonl` or in `llm_comparison/few_shot.txt`.

TABLE 20 — TEST ITEMS 097–120

#	BRAILLE INPUT / ENGLISH REFERENCE	B	C	CER
097	⠠⠙⠊⠑ ⠞⠕⠉⠞⠕⠗ ⠕⠏⠑⠝⠑⠙ ⠙⠁⠎ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠁⠇⠕⠝⠛ ⠙⠁⠎ ⠉⠕⠁⠎⠞⠲ The doctor opened the small boat along the coast.	6	1	65.3%
098	⠠⠙⠊⠑ ⠛⠁⠗⠙⠑⠝⠑⠗ ⠑⠭⠁⠍⠊⠝⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠃⠑⠎⠊⠙⠑ ⠞⠓⠑ ⠓⠁⠗⠃⠕⠥⠗⠲ The gardener examined several old books beside the harbour.	2	1	62.7%
099	⠠⠙⠊⠑ ⠁⠗⠞⠊⠎⠞ ⠉⠇⠕⠎⠑⠙ ⠁ ⠺⠑⠞ ⠥⠍⠃⠗⠑⠇⠇⠁ ⠊⠝ ⠙⠊⠑ ⠍⠕⠗⠝⠊⠝⠛⠲ The artist closed a wet umbrella in the morning.	1	2	70.8%
100	⠠⠙⠊⠑ ⠋⠁⠗⠍⠑⠗ ⠁⠗⠗⠁⠝⠛⠑⠙ ⠗⠊⠏⠑ ⠗⠑⠙ ⠁⠏⠏⠇⠑⠎ ⠊⠝ ⠙⠊⠑ ⠍⠕⠗⠝⠊⠝⠛⠲ The farmer arranged ripe red apples in the morning.	4	1	72.5%
101	⠠⠙⠊⠑ ⠞⠁⠊⠇⠕⠗ ⠎⠅⠑⠞⠉⠓⠑⠙ ⠍⠁⠝⠽ ⠺⠕⠕⠙⠑⠝ ⠉⠓⠁⠊⠗⠎ ⠁⠃⠕⠧⠑ ⠞⠓⠑ ⠧⠊⠇⠇⠁⠛⠑⠲ The tailor sketched many wooden chairs above the village.	2	1	63.2%
102	⠠⠙⠊⠑ ⠓⠥⠝⠞⠑⠗ ⠎⠞⠥⠙⠊⠑⠙ ⠎⠑⠧⠑⠗⠁⠇ ⠕⠇⠙ ⠃⠕⠕⠅⠎ ⠁⠋⠞⠑⠗ ⠞⠓⠑ ⠗⠁⠊⠝⠲ The hunter studied several old books after the rain.	1	1	69.2%
103	⠠⠙⠊⠑ ⠍⠑⠗⠉⠓⠁⠝⠞ ⠍⠑⠝⠞⠊⠕⠝⠑⠙ ⠁ ⠓⠑⠁⠧⠽ ⠃⠕⠭ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲ The merchant mentioned a heavy box inside the museum.	4	2	64.2%
104	⠠⠙⠊⠑ ⠺⠑⠁⠧⠑⠗ ⠏⠁⠊⠝⠞⠑⠙ ⠁ ⠏⠁⠏⠑⠗ ⠍⠁⠏ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲ The weaver painted a paper map inside the museum.	6	1	73.5%
105	⠠⠙⠊⠑ ⠞⠑⠁⠉⠓⠑⠗ ⠏⠁⠊⠝⠞⠑⠙ ⠁ ⠏⠁⠏⠑⠗ ⠍⠁⠏ ⠃⠑⠓⠊⠝⠙ ⠞⠓⠑ ⠃⠁⠗⠝⠲ The teacher painted a paper map behind the barn.	6	2	70.8%
106	⠠⠙⠊⠑ ⠇⠊⠃⠗⠁⠗⠊⠁⠝ ⠏⠗⠑⠏⠁⠗⠑⠙ ⠁ ⠺⠑⠞ ⠥⠍⠃⠗⠑⠇⠇⠁ ⠞⠓⠗⠕⠥⠛⠓⠕⠥⠞ ⠞⠓⠑ ⠺⠊⠝⠞⠑⠗⠲ The librarian prepared a wet umbrella throughout the winter.	2	5	60.0%
107	⠠⠙⠊⠑ ⠋⠁⠗⠍⠑⠗ ⠝⠕⠞⠊⠉⠑⠙ ⠍⠁⠝⠽ ⠺⠕⠕⠙⠑⠝ ⠉⠓⠁⠊⠗⠎ ⠝⠑⠁⠗ ⠞⠓⠑ ⠗⠊⠧⠑⠗⠲ The farmer noticed many wooden chairs near the river.	4	2	66.0%
108	⠠⠙⠊⠑ ⠑⠝⠛⠊⠝⠑⠑⠗ ⠎⠞⠥⠙⠊⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠃⠑⠓⠊⠝⠙ ⠞⠓⠑ ⠃⠁⠗⠝⠲ The engineer studied some fresh bread behind the barn.	3	2	63.0%
109	⠠⠙⠊⠑ ⠝⠥⠗⠎⠑ ⠏⠗⠑⠏⠁⠗⠑⠙ ⠞⠓⠑ ⠛⠇⠁⠎⠎ ⠺⠊⠝⠙⠕⠺ ⠁⠋⠞⠑⠗ ⠞⠓⠑ ⠗⠁⠊⠝⠲ The nurse prepared the glass window after the rain.	1	2	64.7%
110	⠠ ⠎⠉⠊⠑⠝⠞⠊⠎⠞ ⠑⠭⠁⠍⠊⠝⠑⠙ ⠞⠓⠑ ⠞⠁⠇⠇ ⠇⠁⠙⠙⠑⠗ ⠃⠑⠝⠑⠁⠞⠓ ⠞⠓⠑ ⠃⠗⠊⠙⠛⠑⠲ A scientist examined the tall ladder beneath the bridge.	1	15	66.1%
111	⠠⠙⠊⠑ ⠺⠑⠁⠧⠑⠗ ⠗⠑⠏⠁⠊⠗⠑⠙ ⠞⠓⠑ ⠝⠁⠗⠗⠕⠺ ⠗⠕⠁⠙ ⠥⠝⠙⠑⠗ ⠞⠓⠑ ⠕⠇⠙ ⠞⠗⠑⠑⠲ The weaver repaired the narrow road under the old tree.	5	2	61.8%
112	⠠⠙⠊⠑ ⠇⠊⠃⠗⠁⠗⠊⠁⠝ ⠉⠕⠇⠇⠑⠉⠞⠑⠙ ⠁ ⠓⠑⠁⠧⠽ ⠃⠕⠭ ⠃⠑⠋⠕⠗⠑ ⠞⠓⠑ ⠎⠞⠕⠗⠍⠲ The librarian collected a heavy box before the storm.	4	2	43.4%
113	⠠⠙⠊⠑ ⠺⠑⠁⠧⠑⠗ ⠕⠏⠑⠝⠑⠙ ⠞⠓⠑ ⠟⠥⠊⠑⠞ ⠛⠁⠗⠙⠑⠝ ⠥⠝⠙⠑⠗ ⠞⠓⠑ ⠕⠇⠙ ⠞⠗⠑⠑⠲ The weaver opened the quiet garden under the old tree.	1	2	61.1%
114	⠠⠍⠽ ⠝⠑⠊⠛⠓⠃⠕⠥⠗ ⠏⠁⠊⠝⠞⠑⠙ ⠁ ⠃⠗⠊⠛⠓⠞ ⠇⠁⠝⠞⠑⠗⠝ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲ My neighbour painted a bright lantern inside the museum.	3	2	75.0%
115	⠠ ⠎⠁⠊⠇⠕⠗ ⠉⠕⠥⠝⠞⠑⠙ ⠞⠓⠑ ⠑⠍⠏⠞⠽ ⠃⠁⠎⠅⠑⠞ ⠃⠑⠝⠑⠁⠞⠓ ⠞⠓⠑ ⠃⠗⠊⠙⠛⠑⠲ A sailor counted the empty basket beneath the bridge.	1	4	67.9%
116	⠠⠙⠊⠑ ⠋⠊⠎⠓⠑⠗⠍⠁⠝ ⠛⠁⠞⠓⠑⠗⠑⠙ ⠞⠓⠑ ⠛⠗⠑⠑⠝ ⠋⠊⠑⠇⠙ ⠊⠝ ⠞⠓⠑ ⠍⠕⠗⠝⠊⠝⠛⠲ The fisherman gathered the green field in the morning.	3	1	50.0%
117	⠠ ⠙⠁⠝⠉⠑⠗ ⠺⠁⠞⠉⠓⠑⠙ ⠞⠓⠑ ⠑⠍⠏⠞⠽ ⠃⠁⠎⠅⠑⠞ ⠊⠝ ⠞⠓⠑ ⠍⠕⠗⠝⠊⠝⠛⠲ A dancer watched the empty basket in the morning.	1	3	73.5%
118	⠠⠙⠊⠑ ⠉⠁⠏⠞⠁⠊⠝ ⠛⠁⠞⠓⠑⠗⠑⠙ ⠎⠕⠍⠑ ⠋⠗⠑⠎⠓ ⠃⠗⠑⠁⠙ ⠁⠋⠞⠑⠗ ⠞⠓⠑ ⠗⠁⠊⠝⠲ The captain gathered some fresh bread after the rain.	3	1	66.0%
119	⠠⠙⠊⠑ ⠛⠁⠗⠙⠑⠝⠑⠗ ⠎⠞⠁⠗⠞⠑⠙ ⠁ ⠃⠗⠊⠛⠓⠞ ⠇⠁⠝⠞⠑⠗⠝ ⠊⠝⠎⠊⠙⠑ ⠞⠓⠑ ⠍⠥⠎⠑⠥⠍⠲ The gardener started a bright lantern inside the museum.	7	1	57.1%
120	⠠ ⠉⠁⠗⠏⠑⠝⠞⠑⠗ ⠝⠕⠞⠊⠉⠑⠙ ⠞⠓⠑ ⠎⠍⠁⠇⠇ ⠃⠕⠁⠞ ⠁⠗⠕⠥⠝⠙ ⠞⠓⠑ ⠍⠁⠗⠅⠑⠞⠲ A carpenter noticed the small boat around the market.	3	3	71.7%

End of the held-out set. All 120 items are unique, and none appears in training or in the few-shot examples.

19 Running another LLM

SAME 120 ITEMS · SAME PROMPT · SAME SCORER

The kit in `llm_comparison/` is model-agnostic. It contains the inputs, the references, the exact prompt, a scorer, and the three BrailleLM reference rows. A comparison is only meaningful if the conditions are recorded with the score.

TABLE 21 — KIT CONTENTS

FILE	PURPOSE
`test_set.jsonl`	120 items: id, braille, print
`braille_only.txt`	inputs only, in id order
`references.txt`	answers; keep from the model
`prompt.txt`	exact prompt template
`few_shot.txt`	8 train pairs for an 8-shot run
`predictions_template.tsv`	id column + empty answers
`score_llm.py`	scorer and comparison table
`reference_scores.json`	native, baseline, control
`COMPARISON_BRIEF.md`	paste-ready task brief

19.1 CONDITIONS FOR THE HEADLINE ROW

- **Zero-shot.** No fine-tuning, retrieval, or access to `references.txt`.
- **Exact prompt.** Use `prompt.txt`, one braille line per call.
- **Greedy.** Temperature 0 where the API exposes it; otherwise record the setting.
- **Short answer.** Stop at the sentence or 40 new tokens; no explanation in the answer field.
- **One pass.** Report first answers; do not retry individual rows.
- **All 120.** Missing rows are not correct.
- **UTF-8.** Preserve U+2800 spaces and the final period.

19.2 THE PROMPT

```
Translate the following braille to English text.
Braille: {BRAILLE LINE}
English:
```

19.3 SCORE THE ANSWERS

```
cd llm_comparison
python score_llm.py predictions.tsv --name "Model name"
python score_llm.py predictions.jsonl --name "Model name" --out results/
```

The scorer accepts TSV/CSV with `id, text`, JSONL with `{"id", "prediction"}`, or a plain text file in id order with `--order`. It prints native, baseline, control, and the new model in one table, then writes `scores_<name>.json` and `per_row_<name>.csv`.

> THE COMPARISON IS NOT LIKE-FOR-LIKE BY DEFAULT
>
> BrailleLM was fine-tuned on 480 sentences from this exact generator. A general model run zero-shot is being tested on a distribution it has never seen, but with far more parameters and prior language knowledge. Report the condition — zero-shot, 8-shot, or fine-tuned — beside every number. An 8-shot run is not a substitute for fine-tuning, and a fine-tuned external model is not a zero-shot baseline.

19.4 MINIMUM RECORD FOR A COMPARISON

FIELD	RECORD
Model and version	exact API or checkpoint identifier, not a family name
Condition	zero-shot, 8-shot, or fine-tuned; training data and steps if fine-tuned
Decoding	greedy or temperature, top-p, max new tokens, stop sequence
Input and prompt	`test_set.jsonl` (id order) and `prompt.txt` verbatim
Harness	`score_llm.py --name`, date, and the output JSON

20 Score sheet and reading

FILL THE BLANK ROWS · THEN INTERPRET THE PATTERN

A single exact-match number is not a verdict. The pattern across exact match, CER, distinct outputs, vocabulary, and the actor slot says whether a model reads the input or completes a template.

TABLE 22 — COMPARISON SCORE SHEET

MODEL · CONDITION	EXACT	CER	BLEU	DISTINCT	VOCAB	ACTOR
BrailleLM native · fine-tuned · 128 tok	120 / 120	0.0%	100.0	120	138 / 138	100.0%
BrailleLM baseline · fine-tuned · 192 tok	0 / 120	63.8%	4.0	9	22 / 138	2.7%
Baseline 256 control · fine-tuned · 256 tok	0 / 120	60.2%	4.2	15	29 / 138	8.0%
External model · condition	–	–	–	–	–	–
External model · condition	–	–	–	–	–	–
External model · condition	–	–	–	–	–	–

Actor is word accuracy in slot 2 of the 112 nine-word items. Blank rows are for external models; copy the values printed by `score_llm.py`. Do not compare a zero-shot row with a fine-tuned row without saying so.

20.1 READING THE PATTERN

PATTERN	LIKELY MEANING
High exact · high distinct	Reads the input; the useful case
High exact · low distinct	Template memorisation; inspect outputs
Low exact · moderate CER · high BLEU	Near miss or paraphrase; read row errors
Low exact · low distinct · low vocab	Prior only; the baseline failure mode

20.2 NOISE AND COMPARISON

At n = 120, a score near 100% has a 95% Wilson interval of about ±3 points; near 50%, about ±9 points. Two models run on the same 120 items should be compared per row (paired), not by subtracting headline percentages. A gap of 2–3 rows is not a ranking.

Article changes (“A” vs “The”) count as errors, as they do for BrailleLM. If a comparison should treat them as equivalent, score twice and report both conventions.

20.3 THE 256-TOKEN CONTROL’S FIFTEEN PROTOTYPES

01·47	The fisherman prepared the empty basket before the storm.
02·31	The fisherman carried the empty basket before the storm.
03·14	A painter recorded a heavy box before the storm.
04·6	A painter recorded the empty basket before the storm.
05·4	The fisherman carried the iron gate before the storm.
06·4	The fisherman prepared the iron gate before the storm.
07·3	The fisherman prepared the stone bridge under the old tree.
08·3	A painter prepared a wet umbrella before the storm.
09·2	A painter carried a heavy box before the storm.
10·1	A painter recorded the green field before the storm.
11·1	A painter prepared a bright lantern under the old tree.
12·1	A painter recorded a silver coin before the storm.
13·1	A musician prepared a bright lantern before the storm.
14·1	A painter carried a bright lantern before the storm.
15·1	A musician prepared a bright lantern under the old tree.

Fifteen distinct outputs for 120 inputs; the largest covers 47 (39%). The repertoire is still a prior over fragments, not a translation. Full set in `comparison_pack.json` and `preds_baseline_256.tsv`.

> THE COMPARISON TO RUN
>
> Give a general model the 120 braille lines with the prompt above, zero-shot. If it scores high with high diversity, the task is easier for large pretrained models than the BrailleLM result implies. If it scores low with low diversity, the representation result is the interesting one. Either outcome is informative; neither replaces the word-disjoint test in §10.

20.4 REPORTING THE RESULT

A bare percentage is not reproducible. Use one sentence that carries the condition and the reference rows: “Model X, zero-shot, scored 74 / 120 exact, CER 18.3%, BLEU 71.2, 61 distinct outputs, on the BrailleLM held-out set; BrailleLM native scored 120 / 120 and the matched baseline 0 / 120.”

Attach **`scores_model-x.json`** and **`per_row_model-x.csv`** so the row errors can be checked.

21 Appendix

APPENDIX A — THE NINE BASELINE PROTOTYPES

RANK	COUNT	SHARE	OUTPUT
1	36	30.0%	The fisherman closed the stone bridge before the storm.
2	28	23.3%	The captain recorded a silver coin in the morning.
3	18	15.0%	The fisherman prepared the stone bridge before the storm.
4	15	12.5%	The captain opened the glass window before the storm.
5	9	7.5%	The tailor closed the stone bridge before the storm.
6	7	5.8%	The fisherman closed the glass window before the storm.
7	5	4.2%	The captain prepared a silver coin in the morning.
8	1	0.8%	The fisherman carried a silver coin during the festival.
9	1	0.8%	The fisherman carried the stone bridge before the storm.

None of the nine appears verbatim in the training file; all are recombinations of frequent fragments. Frequencies are over the 120 test inputs.

Read as a set, the prototypes enumerate a narrow slice of the corpus: a fisherman or captain performs an action on a bridge, window, or coin, usually before a storm or during a morning or festival. The actor, verb, object, and setting vary, but only across the combinations the training file made frequent. Ninety-seven of the 120 test sentences — 81% — fall into the top four prototypes. That is why the output vocabulary collapses to 22 words while the sentences remain grammatical.

APPENDIX B — TERMS

- **BKFT** — braille-knowledge fine-tuning: initialise each added token from the mean of the base embeddings of its print expansion.
- **UEB** — Unified English Braille. Grade 1 is uncontracted, one cell per letter; Grade 2 uses contractions.
- **CER** — character error rate: Levenshtein distance divided by reference length. 0 means identical strings.
- **PEFT / LoRA** — parameter-efficient fine-tuning with low-rank adapters attached to attention projections.
- **Tokens per cell** — braille tokens consumed divided by Unicode braille cells in the input; lower is more efficient.
- **Prototype** — a complete output sentence that the baseline emits for many different inputs.

PROVENANCE

Report date	`2026-11-02`
Metric source	`preds_baseline.tsv, preds_native.tsv, data/test.jsonl`
Recomputation	`analyze_results.py → report_data.json`
Latency and throughput	measured on the machine in Table 5
Corpus construction	600 unique templated sentences, UEB Grade 1, 600/600 round-trip verified
Baseline control	`max_len 256, same recipe; epoch mean loss 2.317; preds_baseline_256.tsv`
Comparison kit	`llm_comparison/ · build_comparison_pack.py → comparison_pack.json`

C Appendix C · every row, side by side · 1 / 4

All 120 held-out items, one row each. The reference is also the native output; the baseline column is shown in full so every error can be read.

TABLE C1 — ROWS 001–030

#	REFERENCE	BASELINE OUTPUT	NATIVE OUTPUT
001	The gardener started a wet umbrella before the storm.	The fisherman prepared the stone bridge before the storm.	The gardener started a wet umbrella before the storm.
002	A dancer watched a bright lantern along the coast.	The fisherman closed the stone bridge before the storm.	A dancer watched a bright lantern along the coast.
003	The librarian recorded a wet umbrella near the river.	The captain opened the glass window before the storm.	The librarian recorded a wet umbrella near the river.
004	A student watched the empty basket beside the harbour.	The fisherman carried a silver coin during the festival.	A student watched the empty basket beside the harbour.
005	The captain examined the iron gate near the river.	The fisherman closed the stone bridge before the storm.	The captain examined the iron gate near the river.
006	The tailor counted the iron gate before the storm.	The fisherman closed the stone bridge before the storm.	The tailor counted the iron gate before the storm.
007	The pilot examined a heavy box above the village.	The fisherman closed the glass window before the storm.	The pilot examined a heavy box above the village.
008	The merchant arranged the stone bridge under the old tree.	The captain recorded a silver coin in the morning.	The merchant arranged the stone bridge under the old tree.
009	The teacher finished ripe red apples under the old tree.	The captain recorded a silver coin in the morning.	The teacher finished ripe red apples under the old tree.
010	The librarian discovered the tall ladder above the village.	The captain recorded a silver coin in the morning.	The librarian discovered the tall ladder above the village.
011	A painter studied some fresh bread above the village.	The fisherman closed the stone bridge before the storm.	A painter studied some fresh bread above the village.
012	The artist measured warm woollen blankets after the rain.	The captain recorded a silver coin in the morning.	The artist measured warm woollen blankets after the rain.
013	The artist started the broken clock along the coast.	The fisherman closed the stone bridge before the storm.	The artist started the broken clock along the coast.
014	My neighbour mentioned the green field in the morning.	The fisherman prepared the stone bridge before the storm.	My neighbour mentioned the green field in the morning.
015	My neighbour explained some fresh bread near the river.	The fisherman prepared the stone bridge before the storm.	My neighbour explained some fresh bread near the river.
016	The merchant discovered ripe red apples along the coast.	The fisherman closed the stone bridge before the storm.	The merchant discovered ripe red apples along the coast.
017	The traveller recorded ripe red apples throughout the winter.	The captain opened the glass window before the storm.	The traveller recorded ripe red apples throughout the winter.
018	The traveller discovered the iron gate near the river.	The fisherman prepared the stone bridge before the storm.	The traveller discovered the iron gate near the river.
019	A painter arranged the small boat inside the museum.	The fisherman closed the stone bridge before the storm.	A painter arranged the small boat inside the museum.
020	The pilot collected a wet umbrella after the rain.	The fisherman closed the stone bridge before the storm.	The pilot collected a wet umbrella after the rain.
021	The librarian planted warm woollen blankets along the coast.	The captain recorded a silver coin in the morning.	The librarian planted warm woollen blankets along the coast.
022	The doctor counted many wooden chairs across the valley.	The tailor closed the stone bridge before the storm.	The doctor counted many wooden chairs across the valley.
023	A student followed the iron gate beneath the bridge.	The fisherman closed the stone bridge before the storm.	A student followed the iron gate beneath the bridge.
024	A dancer discovered the stone bridge in the morning.	The fisherman closed the stone bridge before the storm.	A dancer discovered the stone bridge in the morning.
025	The farmer mentioned some fresh bread along the coast.	The fisherman prepared the stone bridge before the storm.	The farmer mentioned some fresh bread along the coast.
026	The doctor arranged several old books along the coast.	The fisherman prepared the stone bridge before the storm.	The doctor arranged several old books along the coast.
027	The hunter arranged many wooden chairs during the festival.	The captain recorded a silver coin in the morning.	The hunter arranged many wooden chairs during the festival.
028	A scientist planted several old books near the river.	The fisherman prepared the stone bridge before the storm.	A scientist planted several old books near the river.
029	The tailor started the quiet garden during the festival.	The tailor closed the stone bridge before the storm.	The tailor started the quiet garden during the festival.
030	A shopkeeper polished the empty basket near the river.	The fisherman prepared the stone bridge before the storm.	A shopkeeper polished the empty basket near the river.

Every row in file order. Native is exact on all 120 rows, so its column repeats the reference; the baseline column is the delivered 192-token arm. Metrics: §6–§8.

TABLE C2 — ROWS 031–060

#	REFERENCE	BASELINE OUTPUT	NATIVE OUTPUT
031	The nurse measured the quiet garden near the river.	The captain opened the glass window before the storm.	The nurse measured the quiet garden near the river.
032	The teacher repaired some fresh bread near the river.	The captain opened the glass window before the storm.	The teacher repaired some fresh bread near the river.
033	A scientist remembered the small boat behind the barn.	The fisherman closed the stone bridge before the storm.	A scientist remembered the small boat behind the barn.
034	The doctor finished the green field near the river.	The captain opened the glass window before the storm.	The doctor finished the green field near the river.
035	A dancer noticed several old books across the valley.	The fisherman prepared the stone bridge before the storm.	A dancer noticed several old books across the valley.
036	The writer polished some fresh bread inside the museum.	The captain prepared a silver coin in the morning.	The writer polished some fresh bread inside the museum.
037	The captain followed the tall ladder in the morning.	The fisherman closed the stone bridge before the storm.	The captain followed the tall ladder in the morning.
038	The gardener arranged some fresh bread inside the museum.	The captain recorded a silver coin in the morning.	The gardener arranged some fresh bread inside the museum.
039	The writer protected the stone bridge throughout the winter.	The captain recorded a silver coin in the morning.	The writer protected the stone bridge throughout the winter.
040	A painter finished a silver coin after the rain.	The fisherman closed the stone bridge before the storm.	A painter finished a silver coin after the rain.
041	A dancer started the iron gate inside the museum.	The fisherman closed the glass window before the storm.	A dancer started the iron gate inside the museum.
042	The hunter opened ripe red apples above the village.	The fisherman closed the stone bridge before the storm.	The hunter opened ripe red apples above the village.
043	The traveller measured the empty basket above the village.	The captain recorded a silver coin in the morning.	The traveller measured the empty basket above the village.
044	A sailor carried some fresh bread before the storm.	The fisherman closed the stone bridge before the storm.	A sailor carried some fresh bread before the storm.
045	A painter followed some fresh bread near the river.	The fisherman closed the stone bridge before the storm.	A painter followed some fresh bread near the river.
046	The fisherman finished the tall ladder across the valley.	The captain recorded a silver coin in the morning.	The fisherman finished the tall ladder across the valley.
047	The merchant repaired the glass window before the storm.	The tailor closed the stone bridge before the storm.	The merchant repaired the glass window before the storm.
048	A child explained several old books during the festival.	The fisherman closed the stone bridge before the storm.	A child explained several old books during the festival.
049	The librarian started the broken clock before the storm.	The captain recorded a silver coin in the morning.	The librarian started the broken clock before the storm.
050	The engineer polished a wet umbrella under the old tree.	The fisherman closed the stone bridge before the storm.	The engineer polished a wet umbrella under the old tree.
051	The librarian arranged the glass window in the morning.	The captain prepared a silver coin in the morning.	The librarian arranged the glass window in the morning.
052	The merchant counted the empty basket near the river.	The captain opened the glass window before the storm.	The merchant counted the empty basket near the river.
053	The young girl remembered a wet umbrella across the valley.	The captain recorded a silver coin in the morning.	The young girl remembered a wet umbrella across the valley.
054	The hunter mentioned the glass window beneath the bridge.	The captain recorded a silver coin in the morning.	The hunter mentioned the glass window beneath the bridge.
055	The tailor repaired several old books throughout the winter.	The captain recorded a silver coin in the morning.	The tailor repaired several old books throughout the winter.
056	The tailor closed the stone bridge along the coast.	The captain opened the glass window before the storm.	The tailor closed the stone bridge along the coast.
057	A dancer watched the small boat across the valley.	The fisherman closed the stone bridge before the storm.	A dancer watched the small boat across the valley.
058	The merchant followed the narrow road beside the harbour.	The captain recorded a silver coin in the morning.	The merchant followed the narrow road beside the harbour.
059	A dancer folded the glass window behind the barn.	The fisherman closed the glass window before the storm.	A dancer folded the glass window behind the barn.
060	The librarian noticed the quiet garden throughout the winter.	The captain recorded a silver coin in the morning.	The librarian noticed the quiet garden throughout the winter.

Rows 031–060 continue the same order. Every baseline output is a whole sentence from its nine-prototype repertoire, not a partial translation.

TABLE C3 — ROWS 061–090

#	REFERENCE	BASELINE OUTPUT	NATIVE OUTPUT
061	The traveller examined ripe red apples near the river.	The fisherman prepared the stone bridge before the storm.	The traveller examined ripe red apples near the river.
062	The farmer discovered several old books around the market.	The captain recorded a silver coin in the morning.	The farmer discovered several old books around the market.
063	The captain opened a paper map above the village.	The fisherman closed the glass window before the storm.	The captain opened a paper map above the village.
064	A sailor described many wooden chairs along the coast.	The fisherman closed the stone bridge before the storm.	A sailor described many wooden chairs along the coast.
065	The artist repaired ripe red apples above the village.	The fisherman prepared the stone bridge before the storm.	The artist repaired ripe red apples above the village.
066	The gardener counted the narrow road across the valley.	The captain prepared a silver coin in the morning.	The gardener counted the narrow road across the valley.
067	A shopkeeper protected the stone bridge above the village.	The captain recorded a silver coin in the morning.	A shopkeeper protected the stone bridge above the village.
068	The teacher described the narrow road inside the museum.	The fisherman closed the stone bridge before the storm.	The teacher described the narrow road inside the museum.
069	The traveller arranged a heavy box in the morning.	The fisherman closed the stone bridge before the storm.	The traveller arranged a heavy box in the morning.
070	The captain studied the green field near the river.	The captain opened the glass window before the storm.	The captain studied the green field near the river.
071	A student started the empty basket beneath the bridge.	The fisherman carried the stone bridge before the storm.	A student started the empty basket beneath the bridge.
072	The traveller delivered the broken clock inside the museum.	The captain recorded a silver coin in the morning.	The traveller delivered the broken clock inside the museum.
073	The pilot studied a bright lantern along the coast.	The captain opened the glass window before the storm.	The pilot studied a bright lantern along the coast.
074	The pilot repaired the quiet garden along the coast.	The fisherman closed the stone bridge before the storm.	The pilot repaired the quiet garden along the coast.
075	A carpenter watched some fresh bread during the festival.	The captain recorded a silver coin in the morning.	A carpenter watched some fresh bread during the festival.
076	A musician protected the tall ladder during the festival.	The captain recorded a silver coin in the morning.	A musician protected the tall ladder during the festival.
077	The traveller opened the empty basket around the market.	The tailor closed the stone bridge before the storm.	The traveller opened the empty basket around the market.
078	The librarian carried a wet umbrella inside the museum.	The captain prepared a silver coin in the morning.	The librarian carried a wet umbrella inside the museum.
079	A musician repaired the small boat behind the barn.	The fisherman closed the stone bridge before the storm.	A musician repaired the small boat behind the barn.
080	The baker followed the quiet garden along the coast.	The fisherman closed the stone bridge before the storm.	The baker followed the quiet garden along the coast.
081	The doctor examined the small boat above the village.	The fisherman prepared the stone bridge before the storm.	The doctor examined the small boat above the village.
082	A painter folded a heavy box along the coast.	The fisherman closed the stone bridge before the storm.	A painter folded a heavy box along the coast.
083	The farmer remembered the small boat during the festival.	The captain recorded a silver coin in the morning.	The farmer remembered the small boat during the festival.
084	The merchant carried the glass window around the market.	The tailor closed the stone bridge before the storm.	The merchant carried the glass window around the market.
085	The doctor sketched the stone bridge near the river.	The fisherman closed the stone bridge before the storm.	The doctor sketched the stone bridge near the river.
086	The fisherman measured some fresh bread along the coast.	The tailor closed the stone bridge before the storm.	The fisherman measured some fresh bread along the coast.
087	The gardener discovered a wet umbrella behind the barn.	The fisherman prepared the stone bridge before the storm.	The gardener discovered a wet umbrella behind the barn.
088	A scientist noticed a heavy box behind the barn.	The fisherman closed the stone bridge before the storm.	A scientist noticed a heavy box behind the barn.
089	The fisherman opened the stone bridge throughout the winter.	The captain recorded a silver coin in the morning.	The fisherman opened the stone bridge throughout the winter.
090	The baker described the broken clock behind the barn.	The captain opened the glass window before the storm.	The baker described the broken clock behind the barn.

Rows 061–090. The 256-token control behaves the same way; its fifteen prototypes are listed in §20.3.

TABLE C4 — ROWS 091–120

#	REFERENCE	BASELINE OUTPUT	NATIVE OUTPUT
091	The weaver followed the small boat across the valley.	The captain opened the glass window before the storm.	The weaver followed the small boat across the valley.
092	The old man recorded a bright lantern during the festival.	The captain recorded a silver coin in the morning.	The old man recorded a bright lantern during the festival.
093	The captain recorded the quiet garden across the valley.	The tailor closed the stone bridge before the storm.	The captain recorded the quiet garden across the valley.
094	The young girl started several old books during the festival.	The captain recorded a silver coin in the morning.	The young girl started several old books during the festival.
095	The merchant examined warm woollen blankets beside the harbour.	The tailor closed the stone bridge before the storm.	The merchant examined warm woollen blankets beside the harbour.
096	A sailor measured the stone bridge before the storm.	The fisherman closed the stone bridge before the storm.	A sailor measured the stone bridge before the storm.
097	The doctor opened the small boat along the coast.	The fisherman closed the glass window before the storm.	The doctor opened the small boat along the coast.
098	The gardener examined several old books beside the harbour.	The captain recorded a silver coin in the morning.	The gardener examined several old books beside the harbour.
099	The artist closed a wet umbrella in the morning.	The fisherman closed the stone bridge before the storm.	The artist closed a wet umbrella in the morning.
100	The farmer arranged ripe red apples in the morning.	The captain opened the glass window before the storm.	The farmer arranged ripe red apples in the morning.
101	The tailor sketched many wooden chairs above the village.	The captain recorded a silver coin in the morning.	The tailor sketched many wooden chairs above the village.
102	The hunter studied several old books after the rain.	The fisherman closed the stone bridge before the storm.	The hunter studied several old books after the rain.
103	The merchant mentioned a heavy box inside the museum.	The captain opened the glass window before the storm.	The merchant mentioned a heavy box inside the museum.
104	The weaver painted a paper map inside the museum.	The fisherman closed the glass window before the storm.	The weaver painted a paper map inside the museum.
105	The teacher painted a paper map behind the barn.	The fisherman closed the glass window before the storm.	The teacher painted a paper map behind the barn.
106	The librarian prepared a wet umbrella throughout the winter.	The captain recorded a silver coin in the morning.	The librarian prepared a wet umbrella throughout the winter.
107	The farmer noticed many wooden chairs near the river.	The captain opened the glass window before the storm.	The farmer noticed many wooden chairs near the river.
108	The engineer studied some fresh bread behind the barn.	The fisherman prepared the stone bridge before the storm.	The engineer studied some fresh bread behind the barn.
109	The nurse prepared the glass window after the rain.	The fisherman closed the stone bridge before the storm.	The nurse prepared the glass window after the rain.
110	A scientist examined the tall ladder beneath the bridge.	The fisherman closed the stone bridge before the storm.	A scientist examined the tall ladder beneath the bridge.
111	The weaver repaired the narrow road under the old tree.	The tailor closed the stone bridge before the storm.	The weaver repaired the narrow road under the old tree.
112	The librarian collected a heavy box before the storm.	The captain opened the glass window before the storm.	The librarian collected a heavy box before the storm.
113	The weaver opened the quiet garden under the old tree.	The fisherman closed the stone bridge before the storm.	The weaver opened the quiet garden under the old tree.
114	My neighbour painted a bright lantern inside the museum.	The fisherman prepared the stone bridge before the storm.	My neighbour painted a bright lantern inside the museum.
115	A sailor counted the empty basket beneath the bridge.	The fisherman closed the stone bridge before the storm.	A sailor counted the empty basket beneath the bridge.
116	The fisherman gathered the green field in the morning.	The fisherman prepared the stone bridge before the storm.	The fisherman gathered the green field in the morning.
117	A dancer watched the empty basket in the morning.	The fisherman closed the stone bridge before the storm.	A dancer watched the empty basket in the morning.
118	The captain gathered some fresh bread after the rain.	The fisherman prepared the stone bridge before the storm.	The captain gathered some fresh bread after the rain.
119	The gardener started a bright lantern inside the museum.	The captain prepared a silver coin in the morning.	The gardener started a bright lantern inside the museum.
120	A carpenter noticed the small boat around the market.	The fisherman prepared the stone bridge before the storm.	A carpenter noticed the small boat around the market.

Rows 091–120. End of the comparison. All 120 native outputs equal their references character for character.